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1 Sheaves

PRESHEAVES

Let X be a topological space. A **presheaf** of sets is a functor $\mathcal{F} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Set}$, where $\text{ob}(\mathbf{Open}(X))$ is the set of all open subsets of X , and for any $U, V \in \text{ob}(\mathbf{Open}(X))$,

$$\text{Hom}(U, V) = \begin{cases} \{\iota_{VU}\}, & \text{if } V \subseteq U; \\ \emptyset, & \text{if } V \not\subseteq U. \end{cases}$$

where ι_{UV} denotes the unique inclusion morphism from V to U .

The category **Set** in the definition of a presheaf can be replaced by an arbitrary category $\mathcal{C}$, and the corresponding functor is called a **$\mathcal{C}$ -valued presheaf**.

An element $s \in \mathcal{F}(U)$ is called a **section** of $\mathcal{F}$ over U .

The set $\mathcal{F}(U)$ is called the **set of sections** (or simply the **sections**) over U .

The map $\mathcal{F}(\iota_{VU})$ is called the **restriction map**, and $\mathcal{F}(\iota_{VU})(s)$ is written to be $s|_V$ if $s \in \mathcal{F}(U)$.

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on X . A **morphism** $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a collection of maps $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for each open set $U \subseteq X$, such that whenever $V \subseteq U$, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\mathcal{F}(\iota_{VU})} & \mathcal{F}(V) \\ \downarrow \varphi_U & & \downarrow \varphi_V \\ \mathcal{G}(U) & \xrightarrow{\mathcal{G}(\iota_{VU})} & \mathcal{G}(V) \end{array}$$

commutes. All presheaves on X form a category, denoted by **PSh**(X).

An isomorphism is defined to be a morphism having a two-sided inverse.

Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves. The **presheaf kernel** of φ , **presheaf cokernel** of φ , and **presheaf image** of φ are defined to be the presheaves given by $U \mapsto \ker(\varphi_U)$, $U \mapsto \text{coker}(\varphi_U)$ and $U \mapsto \text{im}(\varphi_U)$, respectively.

Let $\mathcal{F}$ be a presheaf on X . The **stalk** of $\mathcal{F}$ at $x \in X$ is the direct limit

$$\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U).$$

The elements of a stalk $\mathcal{F}_x$ are the **germs** of sections of $\mathcal{F}$ at point x .

Denote $s_x \in \mathcal{F}_x$ to be the image of $s \in \mathcal{F}(U)$ under the canonical morphism of the direct limit.

The **étalé space** of the presheaf $\mathcal{F}$ is the disjoint union of all stalks

$$E(\mathcal{F}) = \coprod_{x \in X} \mathcal{F}_x$$

equipped with the topology generated by the basis of sets $U_s = \{s_x \in \mathcal{F}_x \mid x \in U\}$ for all open sets $U \subseteq X$ and all sections $s \in \mathcal{F}(U)$, where s_x denotes the image of s in the stalk $\mathcal{F}_x$.

There is a canonical projection $\pi : E(\mathcal{F}) \rightarrow X$ defined by $s_x \mapsto x$ for each $s_x \in \mathcal{F}_x$. The topology of the étalé space ensures that π is a continuous map (more precisely, a local homeomorphism).

SHEAVES

A presheaf $\mathcal{F}$ on a topological space X is a **sheaf** if, for any open set $U \subseteq X$ and any open covering $\{V_i\}_{i \in I}$ of U , the following conditions hold:

- (1) **Locality**: For any sections $s, t \in \mathcal{F}(U)$, if $s|_{V_i} = t|_{V_i}$ for all $i \in I$, then $s = t$.
- (2) **Gluing**: For a family of sections $\{s_i \in \mathcal{F}(V_i)\}_{i \in I}$ satisfying $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$ for all $i, j \in I$, there exists a section $s \in \mathcal{F}(U)$ such that $s|_{V_i} = s_i$ for all $i \in I$.

The category of sheaves on X , denoted by $\mathbf{Sh}(X)$, is defined as the full subcategory of $\mathbf{PSh}(X)$ consisting of those presheaves that satisfy both the locality and gluing properties.

Proposition 1.1. *Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves on a topological space X . If φ is an isomorphism, then the induced map $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is an isomorphism for every $x \in X$. If in addition $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then φ is an isomorphism iff φ_x is an isomorphism for all $x \in X$.*

Let $\mathcal{F}$ be a presheaf. The **associated sheaf** (or **sheafification**) of $\mathcal{F}$ is a sheaf $\mathcal{F}^+$ which maps each open set U to the set of maps $s : U \rightarrow E(\mathcal{F})$, such that

- (1) for each $x \in U$, $s(x) \in \mathcal{F}_x$;
- (2) for each $x \in U$, there exists an open neighborhood $V \subseteq U$ of x and a section $t \in \mathcal{F}(V)$ such that $t_y = s(y)$ for all $y \in V$.

There is a canonical morphism of presheaves $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ defined by $\mathcal{F}(U) \mapsto \mathcal{F}^+(U)$.

Proposition 1.2 (Universal property of sheafification). *Let $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ be the canonical morphism. Then for any sheaf $\mathcal{G}$ and any morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, there exists a unique morphism of presheaves $\psi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\varphi = \psi \circ \theta$.*

Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. The **kernel** of φ is defined to be $\ker_{\mathbf{PSh}} \varphi$; the **cokernel** and **image** of φ are defined to be $(\operatorname{coker}_{\mathbf{PSh}} \varphi)^+$ and $(\operatorname{im}_{\mathbf{PSh}} \varphi)^+$, respectively.

The morphism φ is **injective** if $\ker \varphi = 0$. It is **surjective** if $\operatorname{im} \varphi = \mathcal{G}$.

Let $\mathcal{F}'$ be a subsheaf of $\mathcal{F}$. The **quotient sheaf** $\mathcal{F}/\mathcal{F}'$ is defined to be the associated sheaf of the presheaf $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$. Use the definition of associated sheaf, it follows that $(\mathcal{F}/\mathcal{F}')_x = \mathcal{F}_x/\mathcal{F}'_x$.

Let $f : X \rightarrow Y$ be a continuous map of topological spaces.

The **direct image** of a sheaf $\mathcal{F}$ on X is the sheaf $f_*\mathcal{F}$ on Y defined by $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$.

The **inverse image** of a sheaf $\mathcal{G}$ on Y is the sheaf $f^{-1}\mathcal{G}$ on X defined by

$$f^{-1}\mathcal{G}(U) = \left(\varinjlim_{V \supseteq f(U)} \mathcal{G}(V) \right)^+.$$

The étalé space of $f^{-1}\mathcal{G}$ is the pullback $E(\mathcal{G}) \times_Y X$ satisfying the commutative diagram

$$\begin{array}{ccc} E(\mathcal{G}) \times_Y X & \longrightarrow & E(\mathcal{G}) \\ \downarrow & & \downarrow \pi_Y \\ X & \xrightarrow{f} & Y \end{array}$$

Let $\mathcal{F}$ be a sheaf on X and Z is a subset of X . The **restriction** of $\mathcal{F}$ to Z is the inverse image $i^{-1}\mathcal{F}$, where $i : Z \rightarrow X$ is the inclusion map, denote by $\mathcal{F}|_Z$.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Proposition. *Let $\mathcal{F}$ be a presheaf and U be an open set. Then $\varinjlim_{U' \supseteq U} \mathcal{F}(U') \rightarrow \mathcal{F}(U)$ is an isomorphism. If $\mathcal{F}$ is a sheaf, then $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for $V \subseteq U$.*

A sheaf $\mathcal{F}$ is **flabby** (or **flasque**) if for every inclusion $V \subseteq U$ of open sets, $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is surjective.

2 Schemes

RINGED SPACES

A **ringed space** is a pair $(X, \mathcal{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathcal{O}_X : \text{Open}(X)^{\text{op}} \rightarrow \text{Ring}$. The sheaf $\mathcal{O}_X$ is called the **structure sheaf** of $(X, \mathcal{O}_X)$.

A ringed space $(X, \mathcal{O}_X)$ is called a **locally ringed space** if the stalk $\mathcal{O}_{X,x}$ is a local ring for every point $x \in X$.

A **morphism of ringed spaces** from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$ consisting of a continuous map $f : X \rightarrow Y$ and a map $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ of sheaves of rings on Y .

A homomorphism of local rings $\varphi : A \rightarrow B$ is called a **local homomorphism** if $\varphi^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$.

A **morphism of locally ringed spaces** is a morphism $(f, f^\#)$ of ringed spaces such that each induced homomorphism $f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ of local rings is a local homomorphism.

An isomorphism is defined to be a morphism having a two-sided inverse.

For any ring R , $\text{Spec}(R)$ is a ringed space ([Atiyah-Macdonald Exercise 3.24](#)). Consider the inclusion map $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$ which induces a morphism of ringed spaces $f : \text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z}_{(p)})$. However, there is another morphism $g : \text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z}_{(p)})$ defined by $(0) \mapsto (p)$. To exclude such examples, we must restrict our attention to morphisms of locally ringed spaces.

SCHEMES

An **affine scheme** is a locally ringed space $(X, \mathcal{O}_X)$ that is isomorphic (as a locally ringed space) to $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ for some ring R .

A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ admitting an open cover $\{U_i\}$ of X such that each open subscheme $(U_i, \mathcal{O}_X|_{U_i})$ is an affine scheme.

A **morphism of schemes** is a morphism as locally ringed spaces.

Denote $\text{sp}(X)$ to be the underlying topological space of a scheme $(X, \mathcal{O}_X)$.

Let A be a ring. A sheaf of rings $\mathcal{O}_{\text{Spec}(A)}$ is defined as follows. For an open set $U \subseteq \text{Spec}(A)$, Let $\mathcal{O}_{\text{Spec}(A)}(U)$ be the set of functions $s : U \rightarrow \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$, such that

- (1) for each $\mathfrak{p} \in U$, $s(\mathfrak{p}) \in A_{\mathfrak{p}}$;
- (2) for each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ of $\mathfrak{p}$ and an element $a \in A$, such that $s(\mathfrak{q}) = a/f$ for all $\mathfrak{q} \in V$ and some $f \notin \mathfrak{q}$.

PROJECTIVE SCHEMES

Let $S = \bigoplus_{n \geq 0} S_n$ be a graded ring. A **homogeneous multiplicatively closed subset** of S is a multiplicatively closed subset T formed by some homogeneous elements.

A **homogeneous localization** by a homogeneous multiplicatively closed subset T is the degree zero part of the local ring $T^{-1}S$, denote by $(T^{-1}S)_0$.

Let $\mathfrak{p}$ be a homogeneous prime ideal. Denote $S_{(\mathfrak{p})}$ to be the homogeneous localization by a homogeneous multiplicatively closed subset whose elements are not in $\mathfrak{p}$.

Let $f \in S_+ = \bigoplus_{n > 0} S_n$ be a homogeneous element. Denote $S_{(f)}$ to be the set of all elements of the form s/f^k , where $\deg s = k \deg f$.

Let S be a graded ring. Define $\text{Proj}(S)$ to be the set of all homogeneous prime ideals of S that do not contain the irrelevant ideal S_+ . Define a topology on $\text{Proj}(S)$ by taking the closed subsets to be of the form $V(\mathfrak{a})$ for any homogeneous ideal $\mathfrak{a}$.

A sheaf of rings $\mathcal{O}_{\text{Proj}(S)}$ is defined as follows. For an open set $U \subseteq \text{Proj}(S)$, Let $\mathcal{O}_{\text{Proj}(S)}(U)$ be the set of functions $s : U \rightarrow \prod_{\mathfrak{p} \in U} S_{(\mathfrak{p})}$, such that

- (1) for each $\mathfrak{p} \in U$, $s(\mathfrak{p}) \in S_{(\mathfrak{p})}$;
- (2) for each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ of $\mathfrak{p}$ and a homogeneous element $a \in S$, such that $s(\mathfrak{q}) = a/f$ for all $\mathfrak{q} \in V$ and some homogeneous $f \notin \mathfrak{q}$.

Proposition 2.5. *Let S be a graded ring.*

- (a) *For any $\mathfrak{p} \in \text{Proj}(S)$, $\mathcal{O}_{\text{Proj}(S), \mathfrak{p}} \cong S_{(\mathfrak{p})}$.*
- (b) *For any homogeneous element $f \in S_+$, let $D_+(f) = \{\mathfrak{p} \in \text{Proj}(S) \mid f \notin \mathfrak{p}\}$. Then $D_+(f)$ is open in $\text{Proj}(S)$. Furthermore, these open sets cover $\text{Proj}(S)$, and $(D_+(f), \mathcal{O}_{\text{Proj}(S)}|_{D_+(f)}) \cong \text{Spec}(S_{(f)})$.*
- (c) *$(\text{Proj}(S), \mathcal{O}_{\text{Proj}(S)})$ is a scheme.*

MORPHISMS BETWEEN SCHEMES

Let S be a fixed scheme. A **scheme over S** is a scheme X , together with a morphism $X \rightarrow S$.

If X and Y are schemes over S , an **S -morphism** of X to Y is a morphism $f : X \rightarrow Y$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & \swarrow & \\ S & & \end{array}$$

commutes.

Denote **Sch** to be the category of schemes, and **Sch(S)** to be the category of schemes over S .

If R is a ring, denote **Sch(R)** to be the category of schemes over $\text{Spec}(R)$.

Proposition 2.6. *Let k be an algebraically closed field. There is a natural fully faithful functor $t : \mathbf{Var}(k) \rightarrow \mathbf{Sch}(k)$ from the category of varieties over k to schemes over k . For any variety V , its topological space is homeomorphic to the set of closed points of $\text{sp}(t(V))$, and its sheaf of regular functions is obtained by restricting the structure sheaf of $t(V)$ via this homeomorphism.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

The **residue field** of x on a scheme X is the field $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

The **Zariski tangent space** T_x of x on a scheme X is the dual of the $k(x)$ -vector space $\mathfrak{m}_x/\mathfrak{m}_x^2$.

3 First Properties of Schemes

SCHEMES

A scheme is **connected** if its topological space is connected.

In particular, an affine scheme $\text{Spec } R$ is connected iff R has no idempotent elements.

A scheme X is **reduced** if for every open set U , the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

A scheme is **irreducible** if its topological space is irreducible.

A scheme X is **integral** if for every open set $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an integral domain.

Proposition 3.1. *A scheme is integral iff it is both reduced and irreducible.*

A scheme is **locally Noetherian** if it can be covered by open affine subset $\text{Spec}(A_i)$, where each A_i is Noetherian.

A scheme is **Noetherian** if it is locally Noetherian and quasi-compact.

Proposition 3.2. *A scheme is locally Noetherian iff for every open affine subset $U = \text{Spec } A$, A is a Noetherian ring.*

MORPHISMS

A morphism $f : X \rightarrow Y$ is **locally of finite type**, if there exists a covering of Y by open affine subsets $V_i = \text{Spec}(B_i)$, such that for each i , $f^{-1}(V_i)$ can be covered by open affine subsets $U_{ij} = \text{Spec}(A_{ij})$, where each A_{ij} is a finitely generated B_i -algebra.

The morphism f is of **finite type** if in addition for each i , the cover $\{U_{ij}\}$ has a finite subcover.

A morphism $f : X \rightarrow Y$ is **finite**, if there exists a covering of Y by open affine subsets $V_i = \text{Spec}(B_i)$, such that for each i , $f^{-1}(V_i)$ can be covered by open affine subsets $U_{ij} = \text{Spec}(A_{ij})$, where each A_{ij} is a finitely generated B_i -module.

A morphism $f : X \rightarrow Y$ is **quasi-finite**, if

- (i) for every point $y \in Y$, $f^{-1}(y)$ is a finite set.
- (ii) (i) plus $f^{-1}(y)$ is a discrete set.
- (iii) (ii) plus f is of finite type.

Inequivalent definitions

Hartshorne

Grothendieck SGA

Grothendieck EGA

An **open subscheme** of a scheme X is an open subset U with structure sheaf $\mathcal{O}_X|_U$.

An **open immersion** is a morphism $f : X \rightarrow Y$ which induces an isomorphism of X with an open subscheme of Y .

A **closed immersion** is a morphism $f : Y \rightarrow X$ such that f induces a homeomorphism from $\text{sp}(Y)$ to a closed subset of $\text{sp}(X)$, and the induced map $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective.

A **closed subscheme** of a scheme X is an equivalence class of closed immersions, where $f : Y \rightarrow X$ and $f' : Y' \rightarrow X$ are equivalent if there is an isomorphism $i : Y' \rightarrow Y$ such that $f' = f \circ i$.

DIMENSION AND PRODUCTS

The **dimension** of a scheme X is its dimension as a topological space.

If Z is an irreducible closed subset of X , the **codimension** $\text{codim}(Z, X)$ is the supremum of integers n such that there exists a chain

$$Z = Z_0 < Z_1 < \cdots < Z_n$$

of distinct closed irreducible subsets of X . If Y is any closed subset of X , define

$$\mathrm{codim}(Y, X) = \inf_{Z \subseteq Y} \mathrm{codim}(Z, X),$$

where the infimum is taken over all closed irreducible subsets of Y .

Let S be a scheme, and let X, Y be schemes over S . The **fiber product** of X and Y over S is the categorical product of X and Y in $\mathbf{Sch}(S)$, equivalently, the pullback of the morphism $X \rightarrow S$ and $Y \rightarrow S$ in $\mathbf{Sch}$. We denote the fiber product by $X \times_S Y$.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A morphism $f : X \rightarrow Y$ is **quasi-compact** if the inverse image of open quasi-compact subset of Y is quasi-compact on X .

4 Separated and Proper Morphisms

SEPARATED MORPHISMS

Let $f : X \rightarrow Y$ be the morphism of schemes. The **diagonal morphism** is the unique morphism $\Delta : X \rightarrow X \times_Y X$ satisfying $\pi_k \circ \Delta = \text{id}_X$ for $k = 1, 2$, where $\pi_k : X \times_Y X \rightarrow X$ is the canonical projection to the k -th component.

A morphism $f : X \rightarrow Y$ is **separated** if the diagonal morphism Δ is a closed immersion.

A morphism $f : X \rightarrow Y$ is **quasi-separated** if the diagonal morphism Δ is quasi-compact.

A scheme X is **separated over Y** if there exists a separated morphism $f : X \rightarrow Y$. A scheme X is said to be **separated** if it is separated over $\text{Spec } \mathbb{Z}$.

Proposition 4.1. *Any morphism between affine schemes is separated.*

Corollary 4.2. *A morphism $f : X \rightarrow Y$ is separated iff the image of the diagonal morphism is a closed subset of $X \times_Y X$.*

Theorem 4.3 (Valuative criterion of separatedness). *Let $f : X \rightarrow Y$ be a morphism of schemes. Then the following are equivalent:*

- (i) f is separated;
- (ii) f is quasi-separated, and for every valuation ring R with fraction field K , every morphism $\text{Spec } R \rightarrow Y$, and every morphism $\text{Spec } K \rightarrow X$, there exists at most one morphism $h : \text{Spec } R \rightarrow X$ such that the diagram

$$\begin{array}{ccc} \text{Spec } K & \longrightarrow & X \\ \downarrow i & \nearrow h & \downarrow f \\ \text{Spec } R & \longrightarrow & Y \end{array}$$

commutes.

PROPER MORPHISMS

A morphism $f : X \rightarrow Y$ is **universally closed** if for every scheme Z with a morphism $Z \rightarrow Y$, the projection from the fiber product $X \times_Y Z \rightarrow Z$ is a closed map of the underlying topological spaces.

A morphism $f : X \rightarrow Y$ is **proper** if it is separated, of finite type, and universally closed.

Theorem 4.7 (Valuative criterion of properness). *Let $f : X \rightarrow Y$ be a morphism of schemes. Then the following are equivalent:*

- (i) f is proper;
- (ii) f is quasi separated and of finite type, and for every valuation ring R with fraction field K , every morphism $\text{Spec } R \rightarrow Y$, and every morphism $\text{Spec } K \rightarrow X$, there exists a unique morphism $h : \text{Spec } R \rightarrow X$ such that the diagram

$$\begin{array}{ccc} \text{Spec } K & \longrightarrow & X \\ \downarrow i & \nearrow h & \downarrow f \\ \text{Spec } R & \longrightarrow & Y \end{array}$$

commutes.

Let Y be a scheme. The **projective n -space over Y** is $\mathbb{P}_{\mathbb{Z}}^n \times_{\mathrm{Spec} \mathbb{Z}} Y$, denoted by $\mathbb{P}_Y^n$.

A morphism $f : X \rightarrow Y$ of schemes is **projective** if it factors into $X \xrightarrow{i} \mathbb{P}_Y^n \xrightarrow{\pi} Y$, where i is a closed immersion and π is the canonical projection.

A morphism $f : X \rightarrow Y$ of schemes is **quasi-projective** if it factors into $X \xrightarrow{j} X' \xrightarrow{g} Y$, where j is an open immersion and g is a projective morphism.

Theorem 4.9. *A projective morphism of Noetherian schemes is proper. A quasi-projective morphism of Noetherian schemes is of finite type and separated.*

Proposition 4.10. *Let k be an algebraically closed field. The image of the functor $t : \mathbf{Var}(k) \rightarrow \mathbf{Sch}(k)$ is the set of quasi-projective integral schemes over k . The image of the set of projective varieties is the set of projective integral schemes.*

An **abstract variety** is an integral separated scheme of finite type over an algebraically closed field k .

A **complete variety** is an abstract variety which is proper over the field k .

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

5 Sheaves of Modules

SHEAVES OF MODULES

Let $(X, \mathcal{O}_X)$ be a ringed space. A **sheaf of $\mathcal{O}_X$ -modules** is a sheaf $\mathcal{F} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathbf{Ab}$, such that for each open set $U \subseteq X$, the group $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module, and for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is compatible with the module structures via the ring homomorphism $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$.

A **morphism of sheaves of $\mathcal{O}_X$ -modules** is a morphism $\mathcal{F} \rightarrow \mathcal{G}$ of sheaves, such that for each open set $U \subseteq X$, the map $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is a homomorphism of $\mathcal{O}_X(U)$ -modules.

Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}_X$ -modules. The presheaf

$$U \mapsto \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is a sheaf, which is called the **sheaf Hom** , and denote by $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$.

The **tensor product** of $\mathcal{F}$ and $\mathcal{G}$ is the sheaf associated to the presheaf

$$U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U),$$

denote by $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$.

An $\mathcal{O}_X$ -module $\mathcal{F}$ is **free** if it is isomorphic to a direct sum of copies of $\mathcal{O}_X$.

An $\mathcal{O}_X$ -module $\mathcal{F}$ is **locally free** if X can be covered by open sets U for which $\mathcal{F}|_U$ is a free $\mathcal{O}_X|_U$ -module.

The **rank** of $\mathcal{F}$ on an open set U is the number r_U such that $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r_U}$.

A locally free sheaf of rank 1 is called an **invertible sheaf**.

A **sheaf of ideals** on X is a sheaf of modules $\mathcal{I}$ which is a subsheaf of $\mathcal{O}_X$. In other words, for every open set U , $\mathcal{I}(U)$ is an ideal in $\mathcal{O}_X(U)$.

Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces, $\mathcal{F}$ be an $\mathcal{O}_X$ -module and $\mathcal{G}$ be an $\mathcal{O}_Y$ -module. The **direct image** of $\mathcal{F}$ is the sheaf $f_*\mathcal{F}$. The **inverse image** of $\mathcal{G}$ is the sheaf $f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$, denote by $f^*\mathcal{G}$.

COHERENT SHEAVES

Let A be a ring and M be an A -module. The **associated sheaf** of M on $\text{Spec } A$ is a sheaf $\tilde{M}$ which maps each open set U to the set of maps $s : U \rightarrow \coprod_{\mathfrak{p} \in U} M_{\mathfrak{p}}$, such that

- (1) for each $\mathfrak{p} \in U$, $s(\mathfrak{p}) \in M_{\mathfrak{p}}$;
- (2) for each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ of $\mathfrak{p}$, elements $m \in M$ and $f \in A$, such that $m/f = s(\mathfrak{q})$ for all $\mathfrak{q} \in V$ satisfying $f \notin \mathfrak{q}$.

Let $(X, \mathcal{O}_X)$ be a scheme. A sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$ is **quasi-coherent** if X can be covered by open affine subsets $U_i = \text{Spec}(A_i)$, such that for each i there is an A_i -module M_i with $\mathcal{F}|_{U_i} \cong \tilde{M}_i$. We said that $\mathcal{F}$ is **coherent** if furthermore each M_i can be taken to be a finitely generated A_i -module.

Let A be a ring and M be an A -module. The **associated sheaf** of M on $\text{Spec } A$ is a sheaf $\tilde{M}$ which maps each open set U to the set of maps $s : U \rightarrow \coprod_{\mathfrak{p} \in U} M_{\mathfrak{p}}$, such that

- (1) for each $\mathfrak{p} \in U$, $s(\mathfrak{p}) \in M_{\mathfrak{p}}$;
- (2) for each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ of $\mathfrak{p}$, elements $m \in M$ and $f \in A$, such that $m/f = s(\mathfrak{q})$ for all $\mathfrak{q} \in V$ satisfying $f \notin \mathfrak{q}$.

TWISTED SHEAVES

Let S be a graded ring and M be a graded S -module. The **associated sheaf** of M on $\text{Proj } S$ is a sheaf $\tilde{M}$ which maps each open set U to the set of maps $s : U \rightarrow \coprod_{\mathfrak{p} \in U} M_{(\mathfrak{p})}$, such that

- (1) for each $\mathfrak{p} \in U$, $s(\mathfrak{p}) \in M_{\mathfrak{p}}$;
- (2) for each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ of $\mathfrak{p}$, elements $m \in M$ and $f \in A$, such that $m/f = s(\mathfrak{q})$ for all $\mathfrak{q} \in V$ satisfying $f \notin \mathfrak{q}$.

Let S be a graded ring, and let $X = \text{Proj } S$. For any $n \in \mathbb{Z}$, define $\widetilde{S(n)} = \mathcal{O}_X(n)$. $\mathcal{O}_X(1)$ is called the **twisting sheaf** of Serre. For any sheaf of $\mathcal{O}_X$ modules $\mathcal{F}$, the **twisted sheaf** $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$ is denoted by $\mathcal{F}(n)$.

6 Chain Conditions

Let Σ be a partially ordered set. Σ said to be satisfying the **ascending chain condition** (a.c.c) if, for every increasing sequence $x_1 \leq x_2 \leq \dots$ in Σ there exists n such that $x_n = x_{n+1} = \dots$.

Similarly, Σ said to be satisfying the **descending chain condition** (d.c.c) if, for every decreasing sequence $x_1 \geq x_2 \geq \dots$ in Σ there exists n such that $x_n = x_{n+1} = \dots$.

A module satisfying the ascending chain condition is said to be **Noetherian**. A module satisfying the descending chain condition is said to be **Artinian**.

Proposition 6.2. *M is a Noetherian A -module $\Leftrightarrow$ every submodule of M is finitely generated.*

Proposition 6.3. *Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of A -modules. Then*

(i) *M is Noetherian $\Leftrightarrow M'$ and M'' are Noetherian.*

(ii) *M is Artinian $\Leftrightarrow M'$ and M'' are Artinian.*

Proposition 6.5. *Let A be a Noetherian (resp. Artinian) ring, M a finitely generated A -module. Then M is Noetherian (resp. Artinian).*

A **chain** of submodules of a module M is a sequence (M_i) of submodules of M such that

$$M = M_0 \supsetneq M_1 \supsetneq \dots \supsetneq M_n = 0.$$

The **length** of the chain is n . A **composition series** of M is a maximal chain, that is one which no extra submodules can be inserted, or equivalently M_{i-1}/M_i is simple.

Proposition 6.7. *Suppose that M has a composition series of length n . Then every composition series of M has length n , and every chain can be extended to a composition series.*

The length of a composition series of M is called the **length** of M , denoted as $l(M)$.

Proposition 6.10. *For a vector space,*

$$\text{finite dimension} \iff \text{finite length} \iff \text{a.c.c.} \iff \text{d.c.c.}$$

If the conditions are satisfied, length equals dimension.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A topological space X is said to be **Noetherian** if the open subsets of X satisfy the ascending chain condition. Or equivalently, the closed subsets satisfy descending chain condition.

7 Noetherian Rings

Proposition 7.1. *Any homomorphic image of Noetherian ring is Noetherian.*

Proposition 7.2. *Let A be a subring of B ; suppose that A is Noetherian and that B is finitely generated as an A -module. Then B is Noetherian.*

Proposition 7.3. *If A is Noetherian and S is any multiplicatively closed subset of A , then $S^{-1}A$ is Noetherian.*

Theorem 7.5 (Hilbert's basis theorem). *If A is Noetherian, then the polynomial ring $A[x]$ is Noetherian.*

Proposition 7.8. *Let $A \subseteq B \subseteq C$ be rings. Suppose that A is Noetherian, that C is finitely generated as an A -algebra and that C is either (i) finitely generated as a B -module or (ii) integral over B . Then B is finitely generated as an A -algebra.*

Proposition 7.9 (Zariski's lemma). *Let k be a field, E a finitely generated k -algebra. If E is a field then it is a finite algebraic extension of k .*

PRIMARY DECOMPOSITION IN NOETHERIAN RINGS

Proposition 7.11. *In a Noetherian ring every ideal is a finite intersection of irreducible ideals.*

Proposition 7.12. *In a Noetherian ring every irreducible ideal is primary.*

Proposition 7.13. *In a Noetherian ring every ideal has a primary decomposition.*

Proposition 7.14. *In a Noetherian ring every ideal contains a power of its radical.*

Corollary 7.15. *In a Noetherian ring the nilradical is nilpotent.*

Corollary 7.16. *Let A be a Noetherian ring, $\mathfrak{m}$ a maximal ideal of A , $\mathfrak{q}$ any ideal of A . Then the following are equivalent:*

- (i) $\mathfrak{q}$ is $\mathfrak{m}$ -primary;
- (ii) $\sqrt{\mathfrak{q}} = \mathfrak{m}$;
- (iii) $\mathfrak{m}^n \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some $n > 0$.

Proposition 7.17. *Let $\mathfrak{a} \neq (1)$ be an ideal in a Noetherian ring. Then the prime ideals which belong to $\mathfrak{a}$ are precisely the prime ideals which occur in the set of ideals $(\mathfrak{a} : x)$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A subset of a topological space X is **constructible set** if it is a finite union of locally closed sets.

8 Artinian Rings

Proposition 8.1. *In an Artinian ring every prime ideal is maximal.*

Proposition 8.3. *An Artinian ring has only a finite number of maximal ideals.*

Proposition 8.4. *In an Artinian ring the nilradical is nilpotent.*

The **dimension** of A is the supremum of the length of all chains of prime ideals in A .

Proposition 8.5. *A ring A is Artinian $\Leftrightarrow A$ is Noetherian and $\dim(A) = 0$.*

Proposition 8.6. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal. Then exactly one of the following statements is true:*

- (i) $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ for all n ;
- (ii) $\mathfrak{m}^n = 0$ for some n . In this case A is an Artinian local ring.

Proposition 8.7 (Structure theorem of Artinian rings). *An Artinaian ring A is uniquely (up to isomorphism) a finite direct product of Artinian local rings.*

Proposition 8.8. *Let A be an Artinian local ring. Then the following are equivalent:*

- (i) *every ideal in A is principal;*
- (ii) *the maximal ideal $\mathfrak{m}$ is principal;*
- (iii) $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) \leq 1$.

9 Discrete Valuation Rings and Dedekind Domain

Proposition 9.1. *Let A be a Noetherian domain of dimension 1. Then every non-zero ideal $\mathfrak{a}$ in A can be uniquely expressed as a product of primary ideals whose radicals are all distinct.*

DISCRETE VALUATION RINGS

Let K be a field. A **discrete valuation** on K is a mapping v of K^* onto $\mathbb{Z}$ such that

- (1) $v(xy) = v(x) + v(y)$;
- (2) $v(x + y) \geq \min(v(x), v(y))$.

The set consisting of all $x \in K^*$ such that $v(x) \geq 0$ called the **valuation ring** of v .

An integral domain A is a **discrete valuation ring** if there is a discrete valuation v of its field of fractions K such that A is the valuation ring of v .

Proposition 9.2. *Let A be a Noetherian local domain of dimension 1, $\mathfrak{m}$ its maximal ideal. Then the following are equivalent:*

- (i) A is a discrete valuation ring;
- (ii) A is integrally closed;
- (iii) $\mathfrak{m}$ is a principal ideal;
- (iv) $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$;
- (v) Every non-zero ideal is a power of $\mathfrak{m}$;
- (vi) There exists $x \in A$ such that every non-zero ideal is of the form (x^k) for $k \geq 0$.

DEDEKIND DOMAINS

Theorem 9.3. *Let A be a Noetherian local domain of dimension 1. Then the following are equivalent:*

- (i) A is integrally closed;
- (ii) Every primary ideal in A is a prime power;
- (iii) Every local ring $A_{\mathfrak{p}}$ with $\mathfrak{p} \neq 0$ is a discrete valuation ring.

A ring satisfying the conditions is called a **Dedekind domain**.

FRACTIONAL IDEALS

Let A be an integral domain, K its field of fractions. An A -submodule M of K is a **fractional ideal** of A if $xM \subseteq A$ for some $x \neq 0$.

An A -submodule M of K is an **invertible ideal** if there exists a submodule N of K such that $MN = A$. Then $N = (A : M)$, and M is finitely generated fractional ideal.

Proposition 9.6. *For a fractional ideal M , the following are equivalent:*

- (i) M is invertible;

- (ii) M is finitely generated and , for each prime ideal $\mathfrak{p}$, $M_{\mathfrak{p}}$ is invertible;
- (iii) M is finitely generated and , for each maximal ideal $\mathfrak{m}$, $M_{\mathfrak{m}}$ is invertible;

Theorem 9.8. *Let A be an integral domain. Then A is a Dedekind domain $\Leftrightarrow$ every non-zero fractional ideal of A is invertible.*

EXERCISES

1 Sheaves

1.1. Apply the definition on $\mathcal{F}(U)$ and $\mathcal{F}(\iota_{VU})$.

1.2. (a) $\text{im}(\varphi_x) = \text{im}(\mathcal{F}_x \rightarrow \mathcal{G}_x)$, and $(\text{im } \varphi)_x = \varinjlim_{x \ni U} \text{im}(\mathcal{F}(U) \rightarrow \mathcal{G}(U))$. The diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}_x \\ \downarrow \varphi_U & & \downarrow \varphi_x \\ \mathcal{G}(U) & \longrightarrow & \mathcal{G}_x \end{array} \quad \text{can be refined to} \quad \begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}_x \\ \downarrow \varphi_U & & \downarrow \\ \text{im}(\varphi_U) & \longrightarrow & \varinjlim_{x \ni U} \text{im}(\varphi_U) \\ \downarrow & & \downarrow \\ \mathcal{G}(U) & \longrightarrow & \mathcal{G}_x \end{array}$$

Use [Atiyah-Macdonald Exercise 2.15\(i\)](#) to prove $\varinjlim_{x \ni U} \text{im}(\varphi_U) \subseteq \text{im}(\varphi_x)$ and $\varinjlim_{x \ni U} \text{im}(\varphi_U) \supseteq \text{im}(\varphi_x)$. Use

the [exactness of direct limits](#) to prove the ker case. **This is also correct for presheaves.**

(b) $\Leftarrow$: Use (a) to reduce to $(\ker \varphi)_x = 0 \Rightarrow \ker \varphi = 0$. Let $s \in \mathcal{F}(U)$. For each $x \in U$, $s_x = 0$, where $s_x \in \mathcal{F}_x$. By [Atiyah-Macdonald Exercise 2.15\(ii\)](#), there exists $V_x \ni x$ such that $s|_{V_x} = 0$. When x runs over all points of U we have an open cover $\{V_x\}$. Then use the locality axiom.

$\Rightarrow$: Use (a) to reduce to $(\text{im } \varphi)_x = \mathcal{G}_x \Rightarrow \text{im } \varphi = \mathcal{G}$. Let $t \in \mathcal{G}(U)$. For each $x \in U$, $t_x \in \mathcal{G}_x = (\text{im } \varphi)_x$. By [Atiyah-Macdonald Exercise 2.15\(i\)](#), there exist $V_x \ni x$ and $s_x \in (\text{im } \varphi)(V_x)$ such that $(s_x)_x = t_x$. Since $(t|_{V_x})_x = t_x$, [Atiyah-Macdonald Exercise 2.15\(i\)](#) again gives $W_x \subseteq V_x$, $x \in W_x$ such that $s_x|_{W_x} = t|_{W_x}$. Thus, we have a family of sections $\{s_x|_{W_x} \in (\text{im } \varphi)(W_x)\}$ which satisfies $s_x|_{W_x \cap W_y} = t|_{W_x \cap W_y} = s_y|_{W_x \cap W_y}$. Then use the gluing axiom.

Note that the equation $s_x|_{W_x \cap W_y} = s_y|_{W_x \cap W_y}$ uses the injectiveness of $\text{im } \varphi \hookrightarrow \mathcal{G}$ implicitly.

Another approach is to use [Proposition 1.1](#) on $(\ker \varphi)_x = 0$ and $(\text{im } \varphi)_x = \mathcal{G}_x$.

1.3. (a) $\Rightarrow$: Use gluing axiom. $\Leftarrow$: Use [Exercise 1.2\(a\)](#).

(b) Let $\varphi: \mathcal{O}_{\mathbb{C}} \rightarrow \mathcal{O}_{\mathbb{C}^\times}$, $f \mapsto e^f$ be the sheaf morphism of holomorphic functions. Take $U = \mathbb{C}^\times$, then $\varphi_U: \mathcal{O}_{\mathbb{C}}(U) \rightarrow \mathcal{O}_{\mathbb{C}^\times}(U)$ cannot be surjective.

1.4. (a) If $\varphi^+(s_1) = \varphi^+(s_2)$, $\varphi_x^+(s_{1,x}) = \varphi_x^+(s_{2,x})$. We also have $\mathcal{F}_x \cong \mathcal{F}_x^+$ by the definition of sheafification. Then use [Exercise 1.2\(b\)](#).

(b) **$\text{im } \varphi$ is a presheaf but may not be a sheaf.** By (a), $(\text{im } \varphi)^+ \rightarrow \mathcal{G}^+ = \mathcal{G}$ is injective, so $(\text{im } \varphi)^+$ can be identified as a subsheaf of $\mathcal{G}$.

1.5. [Proposition 1.1](#) and [Exercise 1.2\(b\)](#).

1.6. Work on stalks. Use [Exercise 1.2\(a\)](#) to find the kernel. **Or use [Proposition 1.2](#).**

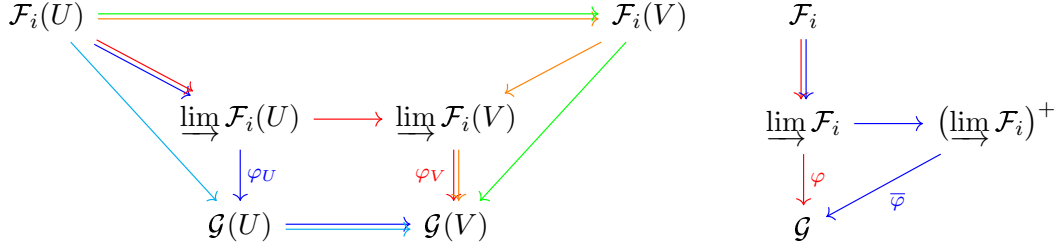
1.7. Use [Exercise 1.2\(a\)](#) to identify the maps on stalks, then use [Proposition 1.1](#). **Or use [Proposition 1.2](#).**

1.8. Denote $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}''$. Use [Exercise 1.2\(b\)](#) $\hookrightarrow$ for injectivity of φ_U .

Let $s \in \mathcal{F}(U)$ such that $\psi_U(s) = 0$. By [Exercise 1.2\(c\)](#), there is $s' \in \mathcal{F}'(U)$ such that $\varphi_x(s'_x) = s_x$. By [Atiyah-Macdonald Exercise 2.15](#) and similar to [Exercise 1.2\(b\)](#) $\Rightarrow$, there is $t_x \in \mathcal{F}'(V_x)$ such that $\varphi_{V_x}(t_x) = s|_{V_x}$. Therefore, $\varphi_{V_x \cap V_y}(t_x|_{V_x \cap V_y}) = s|_{V_x \cap V_y} = \varphi_{V_x \cap V_y}(t_y|_{V_x \cap V_y})$. Due to $\varphi_{V_x \cap V_y}$ is injective, using the gluing axiom of $\mathcal{F}'$, we have $\varphi_{V_x}(t|_{V_x}) = s|_{V_x}$ for some $t \in U$. Since φ is compatible with restriction maps, $\varphi_{V_x}(t|_{V_x}) = (\varphi_U(t))|_{V_x}$. By the locality axiom of $\mathcal{F}$, $\varphi_U(t) = s$.

1.9. Define the canonical projection and the canonical injection.

1.10. $\varphi_U : \varinjlim \mathcal{F}_i(U) \rightarrow \mathcal{G}(U)$ is unique for each U . Follow red $\rightarrow$ orange $\rightarrow$ green $\rightarrow$ cyan $\rightarrow$ blue in the left diagram to prove φ is compatible with the restriction maps, so φ is a morphism of presheaves.



To check it is a sheaf, use [Proposition 1.2](#) to obtain the right diagram.

1.11. Note that [Noetherian spaces are quasi-compact](#). For the locality part, Let $s, t \in \varinjlim \mathcal{F}_i(U)$ and $s|_{V_k} = t|_{V_k}$. By [Atiyah-Macdonald Exercise 2.15\(i\)](#), we have $s_{i_k}|_{V_k} \in \mathcal{F}_{i_k}(V_k)$ and $t_{j_k}|_{V_k} \in \mathcal{F}_{j_k}(V_k)$. Find an index $l \geq i_k, j_k$ for all k due to there are finitely many V_k , so $\mu_l(s|_{V_k}) = \mu_l(t|_{V_k})$. By [Atiyah-Macdonald Exercise 2.15\(ii\)](#), we can find $l' \geq l$ such that $s_{l'}|_{V_k} = t_{l'}|_{V_k}$. Then use the locality of $\mathcal{F}_{l'}(U)$. The gluing part is similar.

1.12. Similar to [Exercise 1.10](#).

1.13. $\Rightarrow$: Let $V_t \subseteq E(\mathcal{F})$ be a basic open set, and let $x \in s^{-1}(V_t)$, then $s(x) \in V_t$. By hypothesis, there exist a neighborhood $W \subseteq U$ of x and $r \in \mathcal{F}(W)$ such that $s(y) = r_y$ for all $y \in W$. Take $W' = W \cap V$. For each $y \in W'$, $r_y = t_y \in V_t$, so W' is an open subset of $s^{-1}(V_t)$.

$\Leftarrow$: Since $s(x) \in \mathcal{F}_x$, by [Atiyah-Macdonald Exercise 2.15\(i\)](#), $s(x) = t_x$ for some $t \in \mathcal{F}(V)$, so V_t is an open neighborhood of $s(x)$, and therefore $s^{-1}(V_t)$ is an open neighborhood of x . For every $y \in s^{-1}(V_t)$, by definition we have $s(y) \in V_t$ and $s(y) \in \mathcal{F}_y$, so $s(y) = t_y$.

1.14. $\mu_U(s) = s_x$. By [Atiyah-Macdonald Exercise 2.15\(ii\)](#), $\mu_{UV}(s) = \mathcal{F}(\iota_{UV})(s) = 0$ for some V .

1.15. Let $\{U_i\}$ be an open cover of U , and let $\varphi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{G}|_{U_i}$ such that $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$. For each $V \subseteq U$, we have $\mathcal{F}|_U(V) = \mathcal{F}(V)$. Let $s \in \mathcal{F}(V)$ and denote $V_i = V \cap U_i$, then $\varphi_i(s|_{V_i}) \in \mathcal{G}|_{U_i}(V_i)$ satisfy $\varphi_i(s|_{V_i})|_{V_i \cap V_j} = \varphi_i|_{V_i \cap V_j}(s|_{V_i \cap V_j}) = \varphi_j(s|_{V_j})|_{V_i \cap V_j}$. Using the gluing property of $\mathcal{G}$ we have a morphism $\varphi_V : \mathcal{F}|_U(V) \rightarrow \mathcal{G}|_U(V)$. Thus there is $\varphi : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$. The locality is similar.

1.16. (b) Denote $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$. Similar to [Exercise 1.8](#), we have $\psi_{V_x \cap V_y}(t_x|_{V_x \cap V_y}) = \psi_{V_x \cap V_y}(t_y|_{V_x \cap V_y})$. Since $\text{im}(\varphi_{V_x \cap V_y}) = \ker(\psi_{V_x \cap V_y})$, there is $r_{xy} \in \mathcal{F}'(V_x \cap V_y)$ such that $\varphi_{V_x \cap V_y}(r_{xy}) = t_x|_{V_x \cap V_y} - t_y|_{V_x \cap V_y}$. By the flasqueness of $\mathcal{F}'$, $\varphi_{V_x}(r_x)|_{V_x \cap V_y} = \varphi_{V_x \cap V_y}(r_{xy})$ for some $r_x \in \mathcal{F}'(V_x)$. Take $t'_x = t_x - \varphi_{V_x}(r_x)$, then t'_x and t_y are compatible on $V_x \cap V_y$.

(c) Use [\(b\)](#) and [blue lemma](#).

(e) If $t \in \mathcal{G}(V)$, let $s|_V = t$ and $s = 0$ at $U \setminus V$. Define $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ to be $s \mapsto (x \mapsto s_x)$.

1.17. Prove that if $y \in \overline{\{x\}}$, then $y \in U$ implies $x \in U$. It is a sheaf due to x has to simultaneously be in or not be in all the open sets which make up an open cover.

1.18. Apply the universal properties of direct limit and of sheafification on the restriction map $\mathcal{F}(f^{-1}(V)) \rightarrow \mathcal{F}(U)$ for $V \supseteq f(U)$. We also have $G(V) \xrightarrow{\mu_V} \varinjlim_{V' \supseteq U'} G(V') \rightarrow \left(\varinjlim_{V' \supseteq U'} G(V') \right)^+$ for $V \supseteq U'$, where $U' = f(f^{-1}(U))$.

1.19. (a) $i^{-1}(V) = V \cap Z$, and $V \cap Z$ is open in the subspace Z .

(b) Use [Proposition 1.1](#) to prove uniqueness.

(c) Work on stalks.

1.20. (a) $\Gamma_{Z \cap U}(U, \mathcal{F}|_U) = \{s \in \mathcal{F}(U) | s_x \neq 0 \text{ for all } x \in Z \cap U\}$. Be aware that we only get $s \in \mathcal{F}(U)$ from the gluing axiom, then use support to derive $s \in \Gamma_{Z \cap U}(U, \mathcal{F}|_U)$.

$$(b) \ j_*(\mathcal{F}|_U)(V) = \left(\varinjlim_{W \supseteq U \cap V} \mathcal{F}(W) \right)^+ = \mathcal{F}(U \cap V).$$

1.21.

1.22. $\mathcal{F}(U) = \{(s_i) \in \prod_{i \in I} \mathcal{F}_i(U \cap U_i) \mid \varphi_{ij}(s_i|_{U \cap U_i \cap U_j}) = s_j|_{U \cap U_i \cap U_j}\}$, $\psi_i : (s_j)_{j \in I} \mapsto s_i$.

2 Schemes

2.1. [Atiyah-Macdonald Exercise 3.21\(i\)](#).

2.2. Let $X = \bigcup_i V_i$ and each $(V_i, \mathcal{O}_X|_{V_i})$ is an affine scheme $\text{Spec}(A_i)$. Since $U = \bigcup_j U_{f_j}$ is the union of some basic open sets, use [Exercise 2.1](#) so $(V_i \cap U_{f_j}, \mathcal{O}_X|_{V_i \cap U_{f_j}}) \cong \text{Spec}((A_i)_{f_j})$.

2.3. (b) $\text{Spec}(A/\mathfrak{A}) \rightarrow \text{Spec}(A)$ is homeomorphic.

(c) The nilradical of $\mathcal{O}_Y(U)$ maps to zero in $\mathcal{O}_X(U)$. Use [universal property of the quotient](#).

2.4. If $X = \text{Spec } B$ is affine, check the induced map $f^\#$ is two-sided inverse of the induced map from $A \rightarrow B$ to $\text{Spec } B \rightarrow \text{Spec } A$, so α is bijective. If $X = \{U_i\}$ is an affine cover, $A \rightarrow \Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(U_i, \mathcal{O}_{U_i})$ induces a map $f_i : U_i \rightarrow \text{Spec } A$. Verify $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and glue them together.

2.5. $\text{Spec } \mathbb{Z} \supseteq (p) \supseteq (0)$. For any ring R , there is a unique map $\mathbb{Z} \rightarrow R$ defined by $1_{\mathbb{Z}} \mapsto 1_R$. Since $\mathcal{O}_{\text{Spec } \mathbb{Z}, x} = \mathbb{Z}_{\mathfrak{p}_x}$, $f_x^\# : \mathcal{O}_{\text{Spec } \mathbb{Z}, f(x)} \rightarrow \mathcal{O}_{X, x}$ is unique by the universal property of localization. $\mathcal{O}_{X, x}$ is a field iff its maximal ideal has a zero preimage, so $f_x^\#$ is local.

2.6. For any ring R , the map $R \rightarrow (0)$ is unique. Note that $\text{Spec}(0) = \emptyset$, so $\mathcal{O}_{(0)}(U) = (0)$, and therefore $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_{(0)}$ is unique.

2.7. $f_x^\# : \mathcal{O}_x \rightarrow K$ is a local homomorphism with $\ker(f_x^\#) = \mathfrak{m}_x$.

2.8. $k[\varepsilon]/(\varepsilon^2)$ has a unique prime ideal (ε) . Let $f_x^\# : \mathcal{O}_{X, x} \rightarrow k[\varepsilon]/(\varepsilon^2)$ be a local homomorphism. The induced map $\overline{f_x^\#} : k(x) \rightarrow k$ is a k -homomorphism, so $k(x) \cong k$. For each $m \in \mathfrak{m}_x$, $f_x^\#(m) = a\varepsilon$ for some $a \in k$, so $f_x^\#$ defines a map $g : \mathfrak{m}_x \rightarrow k$ by $m \mapsto a$. However, $g(\mathfrak{m}_x^2) = 0$, so g induces a map $\bar{g} : \mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow k$.

2.9. $Z \cap U_i$ is irreducible in the subspace U_i . By [Atiyah-Macdonald Exercise 1.20\(iv\)](#), $Z \cap U_i$ has a unique generic point ζ_i . Use the fact that a generic point is contained in every nonempty open subset, and note that $Z \cap U_i \cap U_j$ is open in the subspace $Z \cap U_i$, so $\zeta_i \in Z \cap U_i \cap U_j$. Similarly, $\zeta_j \in Z \cap U_i \cap U_j$, so $\zeta_i = \zeta_j$.

2.10. $\text{Spec } \mathbb{R}$ also contains irreducible $ax^2 + bx + c$.

2.11.

2.12. $X = \{ \coprod X_i \mid (i, x_i) \sim (j, \varphi_{ij}(x_i)) \text{ for } x_i \in U_{ij} \}$, $\psi_i : x_i \mapsto [(i, x_i)]$. Use [Exercise 1.22](#) to obtain the structure sheaf $\mathcal{O}_X(U) = \left\{ (s_i) \in \prod \mathcal{O}_{X_i}(\psi_i^{-1}(U)) \mid \varphi_{ij}^\#(s_i|_{\psi_i^{-1}(U) \cap U_{ij}}) = s_j|_{\varphi_{ij}(\psi_i^{-1}(U) \cap U_{ij})} \right\}$, where $\varphi_{ij}^\# : \mathcal{O}_{X_j}|_{U_{ji}} \rightarrow \mathcal{O}_{X_i}|_{U_{ij}}$ is the induced isomorphism from φ_{ij} .

2.13. [Atiyah-Macdonald Exercise 6.6](#), [Exercise 1.17\(v\)](#), [Exercise 6.8](#).

2.14. (a) $\text{Proj } S = \emptyset \Leftrightarrow S_0 = 0$.

(b) $\varphi(S_+)$ is a homogeneous ideal.

(c) Suppose $\mathfrak{p} \in \text{Proj } T$ but $\mathfrak{p} \not\supseteq \varphi(S_+)$. Let $t \in T_d$ for any $d > 0$. Then we have a sufficient large n such that $nd \geq d_0$, so $t^n \in T_{nd} = \varphi(S_{nd}) \subseteq \mathfrak{p}$, and so $t \in \mathfrak{p}$, which means $T_+ \subseteq \mathfrak{p}$. Prove that $S_{(f)} \rightarrow T_{(\varphi(f))}$ is an isomorphism, then use [Proposition 2.5\(b\)](#).

(d)

2.15.

2.16. (a) Let $\{U_i\}$ be an open affine cover. Then for any $x \in X_f$, there is a $U_i \ni x$ such that $U_i \cap X_f = D(\bar{f})$ is open and contains x .

(b) $a|_{U_i \cap X_f} = 0$. Also $\Gamma(U_i \cap X_f, \mathcal{O}_X|_{U_i \cap X_f}) = B_{f|_{U_i}}$, so $(f^{n_i}a)|_{U_i} = 0$. X is quasi-compact, take $\max(n_i)$, and use the locality axiom.

(c) By (a), $b|_{U_i \cap X_f} = a_i/(f|_{U_i})^{n_i}$ for some $a_i \in \Gamma(U_i \cap X_f, \mathcal{O}_{U_i \cap X_f})$, so $(f|_{U_i})^{n_i} b|_{U_i \cap X_f} = a_i$. Take $n = \max(n_i)$, $(f|_{U_i})^n b|_{U_i \cap X_f} = a'_i$. Therefore, $a'_i|_{U_i \cap U_j \cap X_f} = a'_j|_{U_i \cap U_j \cap X_f}$. Since $U_i \cap U_j$ is quasi-compact, by (b), $(f|_{U_i \cap U_j})^{N_{ij}} (a'_i|_{U_i \cap U_j} - a'_j|_{U_i \cap U_j}) = 0$. Take $N = \max(N_{ij})$, and let $c_i = f^N a'_i \in \Gamma(U_i, \mathcal{O}_X)$, then c_i and c_j are compatible on intersection. Use the gluing axiom there is a $c \in A$ such that $c|_{X_f} = f^{n+N}b$.

(d) Let $\Phi : A_f \rightarrow \Gamma(X_f, \mathcal{O}_{X_f})$, defined by $\frac{a}{f^n} \mapsto \frac{a|_{X_f}}{f^n}$. Φ is injective by (b), and it is surjective by (c).

2.17. (a) Directly from locality and gluing axioms.

(b) By Exercise 2.4, there is a morphism $\varphi : X \rightarrow \text{Spec } A$. By Exercise 2.16(d), $X_{f_i} \rightarrow \text{Spec}(A_{f_i})$ are isomorphisms. By (a), φ is an isomorphism.

2.18. Atiyah-Macdonald Exercise 1.17(ii), Exercise 1.21.

2.19. Atiyah-Macdonald Exercise 1.22.

3 Rings and Modules of Fractions

3.1. The only if part requires the finitely generated property.

3.2. Use ???. Construct the inverse by elementary algebra.

If $M = \mathfrak{a}M$, $S^{-1}M = S^{-1}\mathfrak{a}$, $S^{-1}M = (S^{-1}\mathfrak{a})(S^{-1}M)$. Then use Nakayama's lemma and Exercise 3.1.

3.3. Construct the isomorphism by $(a/s)/(t/1) \mapsto a/(st)$.

3.4. Construct the isomorphism.

3.5. If $a \in A$ is a nilpotent, let $\mathfrak{p} \supseteq \text{Ann}(a)$. $a/1 \in A_{\mathfrak{p}}$ is zero, so $sa = 0$ for some $s \in A - \mathfrak{p}$.

Counter example is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

3.6. Use the fact that for any multiplicatively closed set S , if $A - S$ is an ideal, it's automatically prime.

3.7. (i),(ii) More specifically, $A - \overline{S} = \bigcup_{\mathfrak{p} \cap S = \emptyset} \mathfrak{p}$.

(iii) $\mathfrak{p} \cap (1 + \mathfrak{a}) = \emptyset \Leftrightarrow \mathfrak{a} + \mathfrak{p} \neq (1) \Leftrightarrow \mathfrak{a} + \mathfrak{p} \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. Note that the union of a chain of prime ideals is only a maximum ideal.

3.8. (v) $\Rightarrow$ (ii) If $t/1$ is not a unit in $S^{-1}A$, it is contained in a prime ideal $\mathfrak{q} \subseteq S^{-1}A$, so $t \in T \cap \mathfrak{q}^c$.

3.9. Let S be maximal in Exercise 3.6. If $S_0 \not\subseteq S$, $S \subsetneq SS_0$. Then use Exercise 3.6.

(i) Assume that S_1 is larger, then pick a zero-divisor $x \in S_1$.

(ii) If a/s is not a zero-divisor, a is not a zero-divisor, so $a \in S_0$.

(iii) Note that $s \in S_0$ is a unit.

3.10. (i) ?? says for any element $x \in A$, $x = ax^2$ for some $a \in A$.

(ii) $\Rightarrow$: Use ??. $\Leftarrow$: Consider an $A_{\mathfrak{m}}$ -module. Use ??.

3.11. (i) $\Rightarrow$ (ii) Use ?? on $A/\mathfrak{p}$.

(ii) $\Rightarrow$ (i) Proof that $(A/\mathfrak{A})_{\overline{\mathfrak{p}}}$ has no nilpotent, then use Exercise 1.10 and Exercise 3.10.

(ii) $\Rightarrow$ (iv) Let $f \in \mathfrak{p}_1$ but $f \notin \mathfrak{p}_2$, then $f/1 \in A_{\mathfrak{p}_1}$ is nilpotent, so $sf^n = 0$. We have $\mathfrak{p}_1 \in X_s$ and $\mathfrak{p}_2 \in X_{f^n}$. By Exercise 1.17, $X_s \cap X_{f^n} = X_{sf^n} = \emptyset$.

3.12. (i)-(iii) Use the torsion element definition.

(iv) Use ??.

3.13. Proof they contain in each other. Use ?? for the equivalent conditions.

3.14. Pass to $M/\mathfrak{a}M$. Use ??.

3.15. Let $\{x_i\}_{i=1}^n$ be generators and $\{e_i\}_{i=1}^n$ be the canonical basis. Let $\phi(e_i) = x_i$, so ϕ is surjective. Consider a residue field $k = A/\mathfrak{m}$, we have $\text{Tor}_1(k, F) \rightarrow k \otimes \ker \phi \rightarrow k \otimes F \xrightarrow{1 \otimes \phi} k \otimes F \rightarrow 0$. $\text{Tor}_1(k, F) = 0$ by balancing Tor theorem; $1 \otimes \phi$ bijective so $k \otimes \ker \phi = 0$. By Exercise 2.3, $\ker \phi = 0$.

If $\{x_i\}_{i=1}^m$ generate F where $m < n$, pick some element $\{x_i\}_{i=m+1}^n$ so we also have $\phi(e_i) = x_i$. By injectiveness of ϕ , $\{e_i\}_{i=1}^n$ are linearly dependent.

3.16. (i) $\Rightarrow$ (ii) $f^*(p^e) = p$.

(ii) $\Rightarrow$ (iii) Use ??.

(iii) $\Rightarrow$ (iv) Every module contains a cyclic submodule, which is isomorphic to $A/\mathfrak{a}$. Then use $B \otimes_A A/\mathfrak{a} \cong B/\mathfrak{a}^e$ and $\mathfrak{a}^e \subseteq \mathfrak{m}^e$.

(iv) $\Rightarrow$ (v) Let the mapping be ϕ . Since B flat, $0 \rightarrow (\ker \phi)_B \rightarrow M_B \rightarrow (M_B)_B$ exact. By Exercise 2.13, $M_B \rightarrow (M_B)_B$ injective, so $(\ker \phi)_B = 0$; by assumption, $\ker \phi = 0$.

(v) $\Rightarrow$ (i) Take $M = A/\mathfrak{a}$. Then $M_B = B/\mathfrak{a}^e$.

3.17. If $M \rightarrow N$ injective, we have a diagram

$$\begin{array}{ccc} B \otimes_A M & \xrightarrow{\phi} & B \otimes_A N \\ \downarrow g \otimes \text{id}_M & & \downarrow g \otimes \text{id}_N \\ C \otimes_A M & \xrightarrow{\psi} & C \otimes_A N \end{array}$$

$g \circ f$ flat, so ψ injective. g faithfully flat, by Exercise 3.16(v), $B \otimes_A M \rightarrow C \otimes_B (B \otimes_A M)$ injective. By ??, $C \otimes_B (B \otimes_A M) = C \otimes_A M$, so $g \otimes \text{id}_M$ injective.

3.18. Use Exercise 3.3 to proof $B_{\mathfrak{q}} \cong (B_{\mathfrak{p}})_{\mathfrak{q}B_{\mathfrak{p}}}$. By ?? and ??, $B_{\mathfrak{q}}$ is flat over $B_{\mathfrak{p}}$. Hence $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$ and satisfies Exercise 3.16(iii).

3.19. (v) $M \cong \bigoplus_{i=1}^n A/\mathfrak{a}_i$, so $\text{Supp}(M) = \bigcup_{i=1}^n \text{Supp}(A/\mathfrak{a}_i) = \bigcup_{i=1}^n V(\mathfrak{a}_i) = V(\bigcap_{i=1}^n \mathfrak{a}_i)$. Proof that $\bigcap_{i=1}^n \mathfrak{a}_i = \text{Ann}(M)$.

(vi) Use Exercise 2.3.

(vii) Use Exercise 2.2.

(viii) $\mathfrak{q} \in f^{*-1}(\text{Supp}(M)) \Leftrightarrow M_{\mathfrak{q}^c} \neq 0 \Leftrightarrow M_{\mathfrak{p}} \neq 0$ and $B_{\mathfrak{q}} \neq 0 \Leftrightarrow \mathfrak{q} \in \text{Supp}(M) \cap \text{Supp}(B)$.

3.20. Counter example is $A[x]/(x^2)$.

3.21. (i),(ii) Use ??.

(iii) Use Exercise 1.21(iv).

(iv) To proof $f^{*-1}(\mathfrak{p}) = \text{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})$, use $\mathfrak{q} \in \text{Spec}(S^{-1}B) \Leftrightarrow \mathfrak{q} \cap f(S) = \emptyset$ and $\mathfrak{q} \in \text{Spec}(C/\mathfrak{p}C) \Leftrightarrow f^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$. To proof the equation, note that $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = \text{Frac}(A/\mathfrak{p})$, then apply ?? on $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_{A/\mathfrak{p}} A/\mathfrak{p} \otimes_A B$.

3.22. Let $\mathfrak{p} \in X_f$, so $f \notin \mathfrak{p}$. Then $\mathfrak{q} \in \text{Spec}(A_{\mathfrak{p}}) \Leftrightarrow \mathfrak{q} \subseteq \mathfrak{p} \Leftrightarrow f \notin \mathfrak{q} \Leftrightarrow \mathfrak{q} \in X_f$.

3.23. (i) Use Exercise 1.17(iv).

(v) Define two homomorphisms by the universal properties of direct limit and local ring.

3.24. ?? asserts that there exists a finite open cover (X_{f_i}) such that $\sum_{i=1}^n (f_i) = (f_1, \dots, f_n) = (1)$,

hence $\sqrt{\sum_{i=1}^n (f_i^N)} = \sqrt{\sum_{i=1}^n \sqrt{(f_i^N)}} = (1)$ and therefore $\sum_{i=1}^n (f_i^N) = (1)$ for all N . We can find b_i such that $\sum_{i=1}^n b_i f_i^N = 1$.

Pick N large enough and let $s_i = a'_i/f_i^{m_i} = a_i/f_i^N$. Define $s = \sum_{i=1}^n a_i b_i$. Note that $a_i/f_i^N = a_j/f_j^N$ in $A(U_i \cap U_j)$, and use it to verify s satisfies the condition.

If $t = s - s'$, $t/1 = 0$ in all A_f . Pick f_i are those correspond to the open cover, so $f_i^M t = 0$ for a large M . Then $t = \sum_{i=1}^n b_i f_i^N t = 0$.

3.25. The h is not a homomorphism. Redefine $h(x) = f(x) \otimes 1 = 1 \otimes g(x)$. We have fibers

$$\begin{aligned} f^{*-1}(\mathfrak{p}) &= \operatorname{Spec}(B \otimes_A k(\mathfrak{p})) & g^{*-1}(\mathfrak{p}) &= \operatorname{Spec}(C \otimes_A k(\mathfrak{p})) \\ h^{*-1}(\mathfrak{p}) &= \operatorname{Spec}((B \otimes_A C) \otimes_A k(\mathfrak{p})) \cong \operatorname{Spec}((B \otimes_A k(\mathfrak{p})) \otimes_{k(\mathfrak{p})} (C \otimes_A k(\mathfrak{p}))) \end{aligned}$$

Note that $B \otimes_A k(\mathfrak{p})$ and $C \otimes_A k(\mathfrak{p})$ are vector space, so $\mathfrak{p} \in h^*(T)$ iff $(B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0$, iff $B \otimes_A k \neq 0$ and $C \otimes_A k \neq 0$, iff $\mathfrak{p} \in f^*(Y)$ and $\mathfrak{p} \in g^*(Z)$.

3.26. The fiber $f^{*-1}(\mathfrak{p})$ is the spectrum of $B \otimes_A k(\mathfrak{p})$. By ??, $B \otimes_A k(\mathfrak{p}) = \varinjlim (B_\alpha \otimes_A k(\mathfrak{p}))$. By ??, $f^{*-1}(\mathfrak{p}) = \emptyset \Rightarrow B_\alpha \otimes_A k(\mathfrak{p}) = 0$ for some α , and the reverse is true by the definition of direct limit of rings.

3.27.

3.28.

3.29.

3.30.

4 Primary Decomposition

4.1. Use [Exercise 1.21\(iv\)](#), so $\operatorname{Spec}(A/\mathfrak{a}) = V(\mathfrak{a})$.

4.2. All primary components are prime.

4.3. For any element $x \in A$, $x(1 - ax) = 0 \in \mathfrak{q}$.

4.4. $\mathfrak{m} = \sqrt{\mathfrak{q}} \supsetneq \mathfrak{q} \supsetneq \mathfrak{m}^2$.

4.5. $\mathfrak{m}^2$ is $\mathfrak{m}$ -primary.

4.6. Use Urysohn's lemma to proof that if $f(x_1) = f(x_2) = 0$ for all $f \in \mathfrak{q}$, $\mathfrak{q}$ cannot be primary, so there are infinite many minimal prime ideals. Now use ?? to proof that $(0) = \sqrt{(0)} = \bigcap \mathfrak{p}$ is the intersection of ALL minimal prime ideals.

4.7. (ii),(iv),(v) $(\mathfrak{a}[x])^c = \mathfrak{a}$.

(iii) Let $\bar{f}, \bar{g} \in (A/\mathfrak{q})[x]$. If $\bar{f}\bar{g} = 0$ but $\bar{g} \neq 0$, by ??, $\bar{a}\bar{f} = 0$, so $aa_i \in \mathfrak{q}$. Hence $a_i \in \mathfrak{p}$ and $a_i^{n_i} \in \mathfrak{q}$, so $\bar{a}_i^{n_i} = 0$ and use [Exercise 1.2\(b\)](#).

4.8. $k[x_1, \dots, x_n]/\mathfrak{p}_i = k[x_{i+1}, \dots, x_n]$ integer domain, so $\mathfrak{p}_i$ prime. Note that $\mathfrak{p}_i^n$ is the set of all homogeneous polynomials in $k[x_1, \dots, x_i]$ of degree $\geq n$.

4.9. (i) $\Leftarrow$: x is contained in all prime ideals containing $(0 : a)$, so $x \in \sqrt{(0 : a)}$.

(ii) If $\mathfrak{a} \subseteq \mathfrak{p}$ but $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{p}' \subsetneq S^{-1}\mathfrak{p}$, then $\mathfrak{a} \subseteq \mathfrak{a}^{ec} \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$. If $\mathfrak{b} \subseteq S^{-1}\mathfrak{p}$ but $\mathfrak{b}^c \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$, use $\mathfrak{b}^{ce} = \mathfrak{b}$ derived from ??. $S^{-1}\mathfrak{p}$ is a prime ideal if S doesn't meet $\mathfrak{p}$.

(iii) $(0 : a) = \bigcap (\mathfrak{q}_i : a) \subseteq \mathfrak{p}$. By ??, $(\mathfrak{q}_i : a) \subseteq \mathfrak{p}$ for some i . Then use ?? and the minimality of $\mathfrak{p}$.

4.10. (iv) Let $x \neq 0$, $(0 : x) \subseteq \mathfrak{p}$ for some $\mathfrak{p}$. If $x \in S_{\mathfrak{p}}(0)$, then $sx = 0$, so $s \in (0 : x)$.

4.11. Note that $x \in S(\mathfrak{a}) \Leftrightarrow$ there exists an $s \in S$ such that $sx \in \mathfrak{a}$. Show that $S_{\mathfrak{p}}(0)$ is contained in all $\mathfrak{p}$ -primary ideals.

If $0 = \bigcap \mathfrak{q}_i$ is irredundant, $\sqrt{(0)}$ is the intersection of minimal prime ideals.

4.12. Use ??. Use [Theorem 4.9](#) to show there are at most 2^n different $S(\mathfrak{a})$.

4.13. (i) Use [Exercise 4.12](#).

(ii) Use ?? and [Theorem 4.9](#).

(iii) Proof that $S_{\mathfrak{p}}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = \mathfrak{p}^{(m+n)}$.

(iv) $\Leftarrow$: Show that $\mathfrak{p}^{(n)}$ is the smallest $\mathfrak{p}$ -primary ideal containing $\mathfrak{p}^n$.

4.14. Let $x \notin \mathfrak{q}$. If $ab \in (\mathfrak{q} : x)$ but $a \notin (\mathfrak{q} : x)$, we have $xab \in \mathfrak{q}$ but $xa \notin \mathfrak{q}$. Thus, $b \in (\mathfrak{q} : xa) = (\mathfrak{q} : x)$ by maximality, and therefore $(\mathfrak{q} : x)$ is prime.

4.15. Use [Theorem 4.9](#).

4.16. Use ?? and [Theorem 4.9](#).

4.17.

4.18. (i) $\Rightarrow$ (L1) By [Theorem 4.9](#), $S_{\mathfrak{p}}(\mathfrak{a}) = \bigcap \mathfrak{q}_i$. Pick $x \in S_{\mathfrak{p}}$, then $\mathfrak{q}_i = (\mathfrak{q}_i : x)$ by ??.

(i) $\Rightarrow$ (L2) Use [Exercise 4.12](#).

4.19. Similar to the proof of [Exercise 4.11](#).

The goal is to prove that there exists $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ irredundant, so we need to prove $\mathfrak{q}_i \not\supseteq \bigcap_{j \neq i} \mathfrak{q}_j$. Using the fact that $u\mathfrak{v} \subseteq \mathfrak{q}$ but $u \not\subseteq \mathfrak{q} \Rightarrow \mathfrak{v} \subseteq \mathfrak{p}$, we can assume $\mathfrak{b} = \bigcap_{i=1}^{n-1} \mathfrak{q}_i$ irredundant and $\mathfrak{p}_n$ maximal among $\{\mathfrak{p}_i\}$, so we only need to find $\mathfrak{q}_n \not\supseteq \mathfrak{b}$. If such $\mathfrak{q}_n$ doesn't exist, $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$. Pick a minimal prime ideal $\mathfrak{p} \subseteq \mathfrak{p}_n$ and use [Exercise 4.10](#) and ??.

4.20.

4.21.

4.22.

4.23.

5 Integral Dependence and Valuations

5.1. $f^*(V(I)) = \{f^{-1}(\mathfrak{q}) : \mathfrak{q} \supseteq I\}$. Use ?? to prove $f^*(V(I)) \supseteq V(f^{-1}(I))$.

5.2. $\mathfrak{p} = \ker f$ is prime. By ??, there is a prime ideal $\mathfrak{q} \subseteq B$. $\text{Frac}(B/\mathfrak{q})$ is an algebraic extension of $\text{Frac}(A/\mathfrak{p})$, so $\text{Frac}(A/\mathfrak{p}) \rightarrow \Omega$ can be extended to $\text{Frac}(B/\mathfrak{q}) \rightarrow \Omega$.

5.3. Use the operator $- \otimes_A c^n$.

5.4. If $\left(\frac{1}{x+1}\right)^n + \sum \frac{f_i(x^2-1)}{g_i(x^2-1)(x+1)^i} = 0$, multiply both side by $(x+1)^n$ and take $x = -1$ to get contradiction.

5.5. (ii) Use ?? to prove that the contraction of maximal ideals in B are maximal in A .

5.6. Let $b_i \in B_i$, $f_i \in A[x_i]$ monic, $f_i(b_i) = 0$. Let $f = (f_1, \dots, f_n)$, then $f(b_1, \dots, b_n) = 0$.

5.7. Let $x \in B$ and x integral over A . Let $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$ is the equation of the least degree. Analyze $x(x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1) = -a_0$.

5.8. Prove that there is a ring $D \supseteq B$ such that f and g factor into linear factors in D . Use Kronecker's construction.

5.9. Let $f^n + g_{n-1}f^{n-1} + \dots + g_0 = 0$. Let $f_1 = f - x^r$ with r larger than all $\deg g_i$, then $f_1(f_1^{n-1} + h_{n-1}f_1^{n-2} + \dots + h_1) = -h_0$ with h_0 monic. Then use [Exercise 5.8](#).

5.10. (a) $\Rightarrow$ (b) Let $f^*(V(\mathfrak{q}_1)) = V(I)$, so $I \subseteq \mathfrak{p}_1 \subseteq \mathfrak{p}_2$, and we have $\mathfrak{p}_2 \in V(I) = f^*(V(\mathfrak{q}_1))$.

(a') $\Rightarrow$ (c') By [Exercise 3.23\(v\)](#) and [Exercise 3.26](#), $f^*(\text{Spec}(B_{\mathfrak{q}})) = f^*(\varinjlim \text{Spec}(B_t)) = \bigcap_t f_t^*(\text{Spec}(B_t))$. Each $f_t^*(\text{Spec}(B_t))$ is an open neighborhood of $\mathfrak{p}$, so it contains $A_{\mathfrak{p}}$.

5.11. Use [Exercise 3.18](#) and [Exercise 5.10](#).

5.12. Note that G has identity element so x is a root of the polynomial $\prod_{\sigma \in G} (t - \sigma(x))$. The coefficients are symmetric polynomials of σ_i . G acting on $S^{-1}A$ is defined to be $\sigma(x/s) = \sigma(x)/\sigma(s)$.

5.13. Let $\mathfrak{q} \in P$. Then $\sigma(\mathfrak{q}) \cap A^G = \mathfrak{q} \cap A^G = \mathfrak{p}$, so $\sigma(\mathfrak{q}) \in P$. Let $\mathfrak{q}_1, \mathfrak{q}_2 \in P$ and $x \in \mathfrak{q}_1$. Then $\prod_{\sigma \in G} \sigma(x) \in \mathfrak{q}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{q}_2$, so $x \in \sigma^{-1}(\mathfrak{q}_2)$, and so $\mathfrak{q}_1 \subseteq \bigcup_{\sigma \in G} \sigma(\mathfrak{q}_2)$. Use ?? and ??.

5.14. Note that $\sigma \in G$ fixes A .

5.15.

5.16. We have a homomorphism $f : k[y_1, \dots, y_r] \hookrightarrow k[x_1, \dots, x_n] \twoheadrightarrow A$. Let $a = (a_1, \dots, a_r) \in L$ corresponds to a maximal ideal $\mathfrak{m}_a = (y_1 - a_1, \dots, y_r - a_r)$. By ??, we can find a prime ideal $\mathfrak{p}_a \subset A$ such that $f^{-1}(\mathfrak{p}_a) = \mathfrak{m}_a$, which is contained in a maximal $\mathfrak{n}_a$. Notice that $\mathfrak{n}_a$ correspond a point in X .

5.17. Consider $k[t_1, \dots, t_n]/I(X)$. Use [Exercise 5.16](#).

5.18 (Zariski's lemma). Let n be the number of generators. If $n = 1$, x_1 is invertible, so $x_1^{-1} = \sum_{i=1}^n c_i x_1^i$.

If $n > 1$, let $A = k[x_1]$ and $K_A = \text{Frac}(A)$. Then $K_A[x_2, \dots, x_n]$ is a finite algebraic extension of K_A , so there are monic polynomials $f_i \in K_A[y]$ such that $f_i(x_i) = 0$ for $2 \leq i \leq n$. Pick $a \in A$ such that $a f_2 \cdots f_n \in A[y]$, then $x_2, \dots, x_n$ are integral over A_a . Therefore, $K_A \subseteq B$ are integral over A_a .

If x_1 transcendental over k , A is a UFD. Since every UFD is integrally closed, A is integrally closed, so A_a is integrally closed by ??. K_A integral over A_a , so $A_a = K_A$, but such a must be a unit in A .

5.19. By [Exercise 5.18](#), $k[t_1, \dots, t_n]/\mathfrak{m}$ is a finite algebraic extension of k , so $k[t_1, \dots, t_n]/\mathfrak{m} \cong k$. Let $t_i \mapsto a_i$, then $t_i - a_i \in \mathfrak{m}$. Proof that $(t_1 - a_1, \dots, t_n - a_n)$ is maximal.

5.20. Let $S = A - \{0\}$. By Noether normalization lemma, $S^{-1}B$ is integral over $S^{-1}A[y_1, \dots, y_r]$ with $y_1, \dots, y_r \in S^{-1}B$. The denominators of $y_1, \dots, y_r$ can be absorbed to $S^{-1}A$, so we can assume $y_1, \dots, y_r \in B$. Let $B = A[x_1, \dots, x_n]$, then $x_i/1 \in S^{-1}B$ is integral over $S^{-1}A[y_1, \dots, y_r]$. Reduce the coefficients of all integral dependence equations to a common denominator s , we have $x_i/1$ is integral over $A_s[y_1, \dots, y_r]$. Therefore, $B_s = A_s[x_1/1, \dots, x_n/1]$ is integral over $A_s[y_1, \dots, y_r]$.

5.21. We have

$$\begin{array}{ccccccc} A & \xrightarrow{\phi_A} & A_s & \xrightarrow{\quad} & A_s[y_1, \dots, y_n] & \longrightarrow & B_s \xleftarrow{\phi_B} B \\ & & & \searrow f & \searrow f_s & \searrow f'_s & \downarrow g \\ & & & & & & \Omega \end{array}$$

By [Exercise 5.20](#), B_s integral over $A_s[y_1, \dots, y_n]$, so g is obtained from [Exercise 5.2](#). Then, f_s by universal property of localization; f'_s by defining $y_i \mapsto 0$.

5.22. Let $v \neq 0$ an element of B . Then B_v is integral over A , so $f : A \rightarrow \Omega$ can be extended to $\bar{f} : B_v \rightarrow \Omega$ by [Exercise 5.21](#), with $s \neq 0$ an element of A such that $f(s) \neq 0$. Find an $\mathfrak{m}$ such that $s \notin \mathfrak{m}$ and use [Exercise 5.21](#) again, $g : A \rightarrow \overline{A/\mathfrak{m}}$ can be extended to $\bar{g} : B_v \rightarrow \overline{A/\mathfrak{m}}$. Since $\bar{g}(s) \neq 0$, $\bar{g}$ is not trivial, so $\bar{g}(1/v)\bar{g}(v) = \bar{g}(1) = 1$ means $\bar{g}(v) \neq 0$.

Since $\overline{A/\mathfrak{m}}$ integral over $A/\mathfrak{m}$, $\bar{g}(B_v)$ integral over $A/\mathfrak{m}$, so $\bar{g}(B_v) \cong B_v/\ker(\bar{g})$ is a field and therefore $\ker \bar{g}$ is maximal. By ??, $\ker \bar{g} \cap B$ is maximal in B that doesn't contain v .

5.23. (i) $\Rightarrow$ (ii) Note that $f : A \rightarrow A/\ker f$. Use [Proposition 1.1](#).

(ii) $\Rightarrow$ (iii) Consider $A/\mathfrak{p}$.

(iii) $\Rightarrow$ (i) If $\mathfrak{p}$ is not an intersection of maximal ideals, the Jacobson radical $\mathfrak{J}$ of $A/\mathfrak{p}$ is not zero. Pick $f \in \mathfrak{J}$, then $(A/\mathfrak{p})_f \neq 0$. The contraction of a maximal ideal of $(A/\mathfrak{p})_f$ is $\mathfrak{q} \in A$ which is not maximal and doesn't contain f , but all prime ideals strictly containing $\mathfrak{q}$ contains f .

5.24. (i) Use [Exercise 5.23\(iii\)](#) and going-up theorem.

(ii) Consider $B/\mathfrak{q}$ and $A/\mathfrak{q}^c$. Then use [Exercise 5.22](#), [Exercise 5.23\(i\)](#) and [Proposition 1.1](#).

5.25. (i) $\Rightarrow$ (ii) Similar to the proof of [Exercise 5.22](#), we can find $\bar{g} : B \rightarrow \overline{A/\mathfrak{m}}$. Let $B = A[x_1, \dots, x_n]$, then $\bar{g}(x_i)$ integral over $A/\mathfrak{m}$, and therefore $\bar{g}(B)$ is finitely generated as an $A/\mathfrak{m}$ -algebra. By [Zariski's lemma](#), $\bar{g}(B)$ is finite over $A/\mathfrak{m}$. Note that B is a field, so $\bar{g}$ injective.

(ii) $\Rightarrow$ (i) Take $B = A/\mathfrak{p}$. For any $f \neq 0$ in B , B_f is a finitely generated A -algebra. If B_f is a field, it is finite over A , so integral over A . The integral dependence equation also asserts that B_f is integral over B , so B is a field by ???. If B_f is not a field, it has a non-zero prime ideal whose contraction in B doesn't contain f . Therefore, the intersection of all prime ideals in B is zero.

5.26. (2) $\Rightarrow$ (3) $U_1 \cap X_0 = U_2 \cap X_0 \Rightarrow U_1^c \cap X_0 = U_2^c \cap X_0$, then take closure.

(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) Consider $V(\mathfrak{a}) \cap X_f \cap \text{Max}(A)$. If all maximal ideals in $V(\mathfrak{a})$ contains f , $V(\mathfrak{a}) \cap X_f = \emptyset$ by [Exercise 5.23\(iii\)](#).

(iii) $\Rightarrow$ (i) If not Jacobson, there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subsetneq \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q}$. Let $f \in \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q} \setminus \mathfrak{p}$, then $X_f \cap V(\mathfrak{p}) = \{\mathfrak{p}\}$.

5.27. $\Rightarrow$: Use ?? to proof maximal elements are valuation rings.

$\Leftarrow$: If $(A, \mathfrak{m}) \subsetneq (A', \mathfrak{m}')$, pick $x \in A' \setminus A$. Then $x^{-1} \in \mathfrak{m}$ because it is not a unit in A .

5.28. (1) $\Rightarrow$ (2) Pick $a \in \mathfrak{a} \setminus \mathfrak{b}$, and any $b \in \mathfrak{b}$. Consider a/b and b/a .

(2) $\Rightarrow$ (1) Consider $(a) \subseteq (b)$ or $(b) \subseteq (a)$, we have $a = xb$ or $b = ya$, so $y = x^{-1}$.

5.29. B is a local ring of A means $B = A_{\mathfrak{p}}$ for some prime ideal $\mathfrak{p}$. Let $\mathfrak{m}$ be the maximal ideal of B and let $\mathfrak{p} = \mathfrak{m} \cap A$. If $s \in A$ but $s \notin \mathfrak{p}$, then $s \in B$ and $s \notin \mathfrak{m}$, so s is a unit in B . Let $A_{\mathfrak{p}} \rightarrow B$ to be $x/s \mapsto s^{-1}x$, which is injective; $\mathfrak{p}A_{\mathfrak{p}} \subseteq B$, so B dominates $A_{\mathfrak{p}}$. By [Exercise 5.28](#), $A_{\mathfrak{p}}$ is a valuation ring, then use [Exercise 5.27](#).

5.30. $xy^{-1} \in A$ or $x^{-1}y \in A$, so $\geq$ is a total order. Also $(xz)(yz)^{-1} = xy^{-1}$.

Treat v as a representation map. If $v(x) \geq v(y)$, $(x+y)y^{-1} = xy^{-1} + 1 \in A$.

5.31. Use the given conditions to verify the set is closed under addition and multiplication. Then verify it is a valuation ring by taking $x = y = 1$ and $y = x^{-1}$ in the first condition.

5.32.

5.33.

5.34.

5.35.

6 Chain Conditions

6.1. (i) Let $x \in \ker(u)$. Since u^n surjective, there is a $y \in M$ such that $u^n(y) = x$. But $u^{n+1}(y) = u(x) = 0$, means that $y \in \ker(u^{n+1}) = \ker(u^n)$.

(ii) Let $x \in M$. Then $u^n(x) \in \text{im}(u^n) = \text{im}(u^{n+1})$, so there is a $y \in M$ such that $u^{n+1}(y) = u^n(x)$. But u^n injective, means that $u(y) = x$.

6.2. If M is not Noetherian, there is a submodule N which is not finitely generated. Construct a chain of finitely generated submodules by the generators of N .

6.3. Consider $0 \rightarrow N_2 / (N_1 \cap N_2) \rightarrow M / (N_1 \cap N_2) \rightarrow M/N_2 \rightarrow 0$. Use ??.

6.4. Let $m_1, \dots, m_n$ be generators of M , then $\mathfrak{a} = \bigcap \text{Ann}(m_i)$. $A / \text{Ann}(m_i) \cong Am_i \subseteq M$, then use [Exercise 6.3](#).

6.5. Use the open set definition.

6.6. (i) $\Rightarrow$ (iii) Use [Exercise 6.5](#).

(ii) $\Rightarrow$ (i) Consider a chain of ascending open sets in X .

6.7. Let Σ be the collection of closed subsets of X which are not finite union of irreducible closed subspaces. Then Σ has minimal element X_0 . Proof that X_0 cannot be either irreducible or reducible.

6.8. The converse isn't true. The counter example is $k[x_1, \dots] / (x_1^2, \dots)$.

- 6.9.** The irreducible components of $\text{Spec}(A)$ is $V(\mathfrak{p})$ where $\mathfrak{p}$ minimal (??).
- 6.10.** Use [Exercise 3.19\(v\)](#).
- 6.11.** By [Exercise 6.7](#), $V(\mathfrak{b})$ is a finite union of $V(\mathfrak{q}_i)$, where $\mathfrak{q}_i$ minimal and containing $\mathfrak{b}$. Then use [Exercise 5.10\(c\)](#).
- 6.12.** The converse isn't true. A counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7 Noetherian Rings

7.1 (Cohen's theorem). Let $\mathfrak{a}$ be a maximal element in Σ . If $xy \in \mathfrak{a}$ but $x, y \notin \mathfrak{a}$, then $\mathfrak{a} + (x)$ is finitely generated. Let $\{x, a_1, \dots, a_n\}$ be its generators and let $\mathfrak{a}_0 = (a_1, \dots, a_n)$. Proof that $\mathfrak{a}_0 \not\subseteq \mathfrak{a}$ is impossible. Therefore, if $u \in \mathfrak{a} \subseteq \mathfrak{a} + (x) = \mathfrak{a}_0 + (x)$, $u = a_0 + bx$ with $b \in \mathfrak{a}$, so $\mathfrak{a} \subseteq \mathfrak{a}_0 + x \cdot (\mathfrak{a} : x)$; we also have $x \cdot (\mathfrak{a} : x) \subseteq \mathfrak{a}$. Note that $y \in (\mathfrak{a} : x)$.

7.2. $\Rightarrow$: ??.

$\Leftarrow$: $(a_0, a_1, \dots) \subseteq \mathfrak{R}$ and use [Corollary 7.15](#).

7.3. (i) $\Rightarrow$ (ii) Use ?? and ??.

(ii) $\Rightarrow$ (iii) Let $S = \{1, x, x^2, \dots\}$. Proof that $\bigcup (\mathfrak{a} : x^n) = S(\mathfrak{a})$.

(iii) $\Rightarrow$ (i) Similar to the proof of [Proposition 7.12](#).

7.4. (i) $\mathfrak{p} = \{(z - c) : |c| = 1\}$ is prime, and the ring is $(\mathbb{C}[x])_{\mathfrak{p}}$. Use [Proposition 7.3](#).

(ii) Use ??. If $f = z^n g$ with g invertible, $(f) = (z^n)$, so the only ideals are (0) and (z^n) .

(iii) $\dots \subsetneq (\sin \pi n z) \subsetneq (\sin \pi(n+1)z) \subsetneq \dots$ is an ascending chain.

(iv) The ring is $\mathbb{C}[z^{k+1}]$.

(v) The ring is $\mathbb{C}[z, z^2, \dots, wz, wz^2, \dots, w^2z, w^2z^2, \dots]$.

7.5. Use [Exercise 5.12](#) and [Proposition 7.8](#).

7.6. A ring is finitely generated if it is finitely generated as a $\mathbb{Z}$ -algebra. $\mathbb{Z}$ is a Jacobson ring. By [Exercise 5.25](#), K is finite over $\mathbb{Z}$, so it is of the form $\mathbb{Z}^n \oplus \bigoplus_i \mathbb{Z}/p_i\mathbb{Z}$ due to the addition operation forms a finitely generated abelian group. If $\text{char } K = 0$, $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{Z}^n$, then use [Proposition 7.8](#).

7.7. Use ascending chain condition.

7.8. $A \cong A[x]/(x)$.

7.9. Let $\mathfrak{a}_0 \subseteq \mathfrak{a}_1 \subseteq \dots$ be a chain in A . Since $\mathfrak{a}_0$ is contained in only finite many maximal ideals, there is a large enough n such that $S_{\mathfrak{m}}^{-1}\mathfrak{a}_n = S_{\mathfrak{m}}^{-1}\mathfrak{a}_{n+1} = \dots$ for all $\mathfrak{m} \supseteq \mathfrak{a}_0$. We also have $S_{\mathfrak{m}}^{-1}\mathfrak{a}_i = A_{\mathfrak{m}}$ for all i , when $\mathfrak{m} \not\supseteq \mathfrak{a}_0$. Then apply ?? on $S_{\mathfrak{m}}^{-1}(\mathfrak{a}_{n+i}/\mathfrak{a}_n)$.

7.10. $M[x] = M \otimes A[x]$, so it is a homomorphic image of $M \times A[x]$. Use [Proposition 7.1](#).

7.11. Counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7.12. By [Exercise 3.16\(i\)](#), $\mathfrak{a} \mapsto \mathfrak{a}^e$ is injective. Then use ascending chain condition.

7.13. Transfer the Noetherian property $B \rightarrow B_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ and use [Exercise 6.8](#).

7.14 ([Hilbert's Nullstellensatz](#)). If $f \notin \sqrt{\mathfrak{a}}$, there is a $\mathfrak{p} \supseteq \mathfrak{a}$ such that $f \notin \mathfrak{p}$. $(A/\mathfrak{p})_{\overline{f}}$ is a finitely generated k -algebra, so by [Zariski's lemma](#), $\left((A/\mathfrak{p})_{\overline{f}}\right)/\mathfrak{m} = k$. Let x_i be the images of t_i in $\left((A/\mathfrak{p})_{\overline{f}}\right)/\mathfrak{m}$, then $x = (x_1, \dots, x_n)$ satisfies $x \in V$ but $f(x) \neq 0$.

Rabinowitsch trick: Let $f_1, \dots, f_m$ be generators of $\mathfrak{a}$. If $f \in I(V)$, then $f_1, \dots, f_m, 1 - t_0 f$ has no zeros in $k[t_0, t_1, \dots, t_n]$. Apply [Exercise 5.17](#), $1 = g_0(1 - t_0 f) + \sum_{i=1}^m g_i f_i$. Now put $t_0 = 1/f$, we have $g_i \in k[t_1, \dots, t_n, 1/f]$. Therefore, $g_i = h_i/f^{N_i}$ with $h_i \in k[t_1, \dots, t_n]$. Hence we have $f^N = \sum_{i=1}^m h_i f^{N-N_i} f_i$ for all $N \geq \max_i \{N_i\}$.

7.15. (iv) $\Rightarrow$ (i) Let $x_1, \dots, x_n$ be elements of M whose images in $M/\mathfrak{m}M$ form a basis of k -vector space. Let $N = (x_1, \dots, x_n)$, so $N + \mathfrak{m}M = M$. Since $\mathfrak{J} = \mathfrak{m}$, apply [Nakayama's lemma](#) on M/N , we have $M = N$. Let F be a free A -module with basis $e_1, \dots, e_n$, and $\phi : F \rightarrow M$ be $\phi(e_i) = x_i$. Then

$$\mathrm{Tor}_1^A(k, M) \rightarrow k \otimes_A \ker \phi \rightarrow k \otimes_A F \xrightarrow{1 \otimes \phi} k \otimes_A M \rightarrow 0$$

exact. Since $k \otimes F$ and $k \otimes M = M/\mathfrak{m}M$ are vector spaces of a same dimension, $1 \otimes \phi$ is an isomorphism, so $k \otimes \ker \phi = 0$. By [Exercise 2.3](#), $\ker \phi = 0$.

7.16. Use ?? and [Exercise 7.15](#).

7.17.

7.18.

7.19.

7.20. (i) $\Rightarrow$: We have a chain $\mathcal{F}_0 \subseteq \mathcal{C}_1 \subseteq \mathcal{F}_1 \subseteq \mathcal{C}_2 \subseteq \dots \subseteq \mathcal{F}$, where $\mathcal{F}_0$ is the topology, $\mathcal{C}_i = \mathcal{F}_{i-1} \cup \{S^c : S \in \mathcal{F}_{i-1}\}$, and $\mathcal{F}_i$ is a collection of all finite intersections of sets in $\mathcal{C}_i$.

(ii) $\Rightarrow$: Use the open set definition.

7.21. $\Rightarrow$: Use [Exercise 7.20](#).

$\Leftarrow$: If $E \notin \mathcal{F}$, let $\mathcal{C} = \{X' \subseteq X : X' \text{ closed, } E \cap X' \notin \mathcal{F}\}$. By Noetherian property, $\mathcal{C}$ has a minimal element X_0 . Use the closed set definition to show X_0 is irreducible. If $\overline{E \cap X_0} \neq X_0$, $E \cap \overline{E \cap X_0} \in \mathcal{F}$ by minimality, but $E \cap \overline{E \cap X_0} = E \cap X_0$. If $U_0 \subseteq E \cap X_0$, let $U_0 = U \cap X_0$ where U is open in X , then $E \cap X_0 = U_0 \cup (E \cap U^c \cap X_0)$.

7.22. $\Leftarrow$: If E is not open, let $\mathcal{C} = \{X' \subseteq X : X' \text{ closed, } E \cap X' \text{ is not open}\}$ and use the notations similar to [Exercise 7.21](#). If $U_0 \subseteq E \cap X_0$, $U^c \cap X_0$ is closed, so by minimality $E \cap U^c \cap X_0$ is open.

7.23. Hint.

7.24. Hint.

7.25. Use [Exercise 5.11](#) and [Exercise 7.24](#).

7.26.

7.27.

8 Artinian Rings

8.1. Use [Corollary 7.16](#) on local ring $A_{\mathfrak{p}_i}$. Then apply the contraction. If $\mathfrak{q}_i$ is isolated, then $\mathfrak{p}_i$ is minimal.

8.2. (i) $\Rightarrow$ (ii) Use [Proposition 8.1](#).

(iii) $\Rightarrow$ (i) Use [Proposition 8.5](#).

8.3. (i) $\Rightarrow$ (ii) By [Proposition 8.7](#), we can assume that A is an Artinian local ring. By [Proposition 8.4](#), $\mathfrak{m}^n = 0$ for some n , so we have a chain $A = \mathfrak{m}^0 \supseteq \mathfrak{m}^1 \supseteq \dots \supseteq \mathfrak{m}^n = 0$.

By [Zariski's lemma](#), $A/\mathfrak{m}$ is a finitely generated k -module. By [Proposition 8.5](#), A is Noetherian, so $\mathfrak{m}^i$ is a finitely generated A -module, and therefore $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a finitely generated $A/\mathfrak{m}$ -module. Use

$$\begin{aligned} \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}) &= \dim_{A/\mathfrak{m}}(\mathfrak{m}^i/\mathfrak{m}^{i+1}) \dim_k(A/\mathfrak{m}); \\ \dim_k(\mathfrak{m}^i) &= \dim_k(\mathfrak{m}^{i+1}) + \dim_k(\mathfrak{m}^i/\mathfrak{m}^{i+1}), \end{aligned} \quad \text{where } 0 \rightarrow \mathfrak{m}^{i+1} \rightarrow \mathfrak{m}^i \rightarrow \mathfrak{m}^i/\mathfrak{m}^{i+1} \rightarrow 0$$

to finish the proof.

(ii) $\Rightarrow$ (i) A is a k -vector space of finite dimension. Use [Proposition 6.10](#).

8.4. (i) $\Rightarrow$ (iii) The finitely-generated property is preserved for localization and quotient of modules.

(ii) $\Rightarrow$ (iii) $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is a finitely generated $k(\mathfrak{p})$ -algebra. By [Hilbert's basis theorem](#), any finitely generated $k(\mathfrak{p})$ -algebra is Noetherian. Then use [Exercise 8.2](#) and [Exercise 8.3](#).

(iii) $\Rightarrow$ (ii),(iv) $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is Noetherian and integral over the field $k(\mathfrak{p})$. If $\mathfrak{q} \in B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is prime, $(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})/\mathfrak{q}$ is an integral domain, so it is a field by [??](#). Hence, every prime ideal in $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is maximal, which means $\dim(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = 0$. Thus, $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ is Artinian by [Proposition 8.5](#). Then use [Proposition 8.3](#).

A counter example is $f : \mathbb{Q} \hookrightarrow \overline{\mathbb{Q}}$ where $\overline{\mathbb{Q}}$ is an algebraic closure of $\mathbb{Q}$.

8.5. Use Noether normalization lemma and let $\mathfrak{m}$ be a maximal ideal in $k[y_1, \dots, y_r]$. By [Exercise 8.4](#), $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ is finite over $k[y_1, \dots, y_r]/\mathfrak{m} = k$, so by [Exercise 8.3](#) and [Proposition 8.3](#), $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ only has finite many maximal ideals, correspond to finite many points in X .

Let N be the number of generators of A over $k[y_1, \dots, y_r]$. The k -dimension of k -vector space $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ is uniformly bounded by N . If $\mathfrak{n}_1, \dots, \mathfrak{n}_M$ are maximal ideals in $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$, by [??](#), $A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}} \rightarrow \prod_{i=1}^M (A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})/\mathfrak{n}_i$ is surjective, so $\dim_k(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}) \geq \sum_{i=1}^M \dim_k((A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})/\mathfrak{n}_i) \geq M$.

8.6. Consider $(A/\mathfrak{q})_{\overline{\mathfrak{p}}}$, which is an Artinian local ring. Use the following claims then [Proposition 6.7](#).

- If A is an Artinian local ring, $\mathfrak{m}$ its only maximal ideal. Then every ideal in A is $\mathfrak{m}$ -primary. Radical is the intersection of the only prime ideal.
- There is a 1-1 correspondence between $\mathfrak{p}$ -primary ideals in A and $S^{-1}\mathfrak{p}$ -primary ideals in $S^{-1}A$. Use [??](#) and [Theorem 4.9](#).

9 Discrete Valuation Rings and Dedekind Domain

9.1. [??](#) asserts that $\dim S^{-1}A$ can only be 0 and 1. If $\dim S^{-1}A = 0$, it is $\text{Frac } A$; if $\dim S^{-1}A = 1$, use [??](#).

By [??](#), $S^{-1}(M^{-1}) = S^{-1}(A : M) = (S^{-1}A : S^{-1}M) = (S^{-1}M)^{-1}$. We also have $S^{-1}(MN) = (S^{-1}M)(S^{-1}N)$, so the map $M \mapsto S^{-1}M$ is a homomorphism. Use [??](#). Note that $(u) \mapsto S^{-1}(u) = (S^{-1}A)u$ maps P into P' , so $H \rightarrow H'$ is surjective.

9.2. Localize and use [??](#) to the final move, we can assume that A is a DVR.

[Proposition 9.2\(iii\)\(v\)](#) assert that $c(f) = (x^n)$ where x is the generator of $\mathfrak{m}$, so $f = x^n f_0$ where f_0 is primitive. Apply [??](#) to f_0, g_0 and note that for any $a \in A$, $ac(f) = c(af)$.

9.3. $\Rightarrow$: $(x_1, x_2) = (x_1)$ if $x_1 x_2^{-1} \in A$, so every ideal is principal. Then use [Proposition 9.2\(iii\)](#).

$\Leftarrow$: If $v(y) \geq v(x)$, $y = (yx^{-1})x$, so every ideal in DVR is of the form $\mathfrak{m}_k = \{z \in A : v(z) \geq k\}$.

9.4. For any non-zero $x \in A$, $x \in \mathfrak{m}^n$ but $x \notin \mathfrak{m}^{n+1}$ for some n . Define a discrete valuation $v(x) = n$.

9.5. By [Exercise 3.13](#) and [Exercise 7.16](#), and notice that torsion free module over a PID is flat.

9.6. From [Theorem 9.3\(iii\)](#), [Proposition 9.2\(v\)](#), and the structure theorem for finitely generated modules over a PID, $M_{\mathfrak{p}} = \bigoplus_{i=1}^N A_{\mathfrak{p}}/\mathfrak{p}^{n_i} A_{\mathfrak{p}} = \bigoplus_{i=1}^N (A/\mathfrak{p}^{n_i})_{\mathfrak{p}}$. Define a homomorphism $\phi : M \rightarrow \bigoplus_{\mathfrak{p}} M_{\mathfrak{p}}$ by $m \mapsto (m/1, \dots)$, so each $\phi_{\mathfrak{p}}$ is an isomorphism. By [??](#), ϕ is an isomorphism.

By [Exercise 3.19\(v\)](#), $\text{Supp}(M) = V(\text{Ann}(M))$. From [Proposition 9.1](#) and [Theorem 9.3\(ii\)](#), $\text{Ann}(M) = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_k}$. Use $\dim A = 1$ to proof that $V(\text{Ann}(M))$ is finite.

9.7. From [Proposition 9.1](#) and [Theorem 9.3\(ii\)](#), $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_k}$. By [??](#), $A \rightarrow \prod_{i=1}^k A/\mathfrak{p}_i^{n_i}$ is a surjective with kernel $\mathfrak{a}$. Note that each component $A/\mathfrak{p}_i^{n_i}$ is local so is PID.

If $a \in \mathfrak{b}$, $\mathfrak{b}/(a) = (\overline{b})$ and therefore $\mathfrak{b} = (a, b)$.

9.8. Localize and note that all ideals in DVR are $\{z \in A : v(z) \geq k\}$. Use [??](#) to pull back.

9.9. This is equivalent to say that

$$A \xrightarrow{\phi} \bigoplus_{i=1}^n A/\mathfrak{a}_i \xrightarrow{\psi} \bigoplus_{i < j} A/(\mathfrak{a}_i + \mathfrak{a}_j)$$

is exact, where $\phi(x) = (x + \mathfrak{a}_i)_{i=1}^n$, and $\psi(x_i + \mathfrak{a}_i)_{i=1}^n = (x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j)_{i < j}$. If A is DVR, we have $\mathfrak{a}_i + \mathfrak{a}_j = \mathfrak{a}_{\min(i,j)}$. If $(x_i + \mathfrak{a}_i)_{i=1}^n \in \ker \psi$, $x_i + \mathfrak{a}_i = x_j + \mathfrak{a}_i$ for all $i < j$, so take $j = n$ for all i and we have $\phi(x_n) \in \ker \psi$. Then use [Theorem 9.3\(iii\)](#) and [??](#) for Dedekind domain.

SUMMARY

Chapter 1 - Sheaves

Atiyah-Macdonald Exercise 2.15 is used thoroughly. (i) means $t \in \mathcal{F}_x$ can be represented by $s \in \mathcal{F}(U)$, (ii) is used for finding agreement on a smaller neighborhood.

The restriction $\mathcal{F}|_Z$ can be simplified if Z is an open set.

Chapter 2 - Schemes

Most exercises help to enhance the comprehension of the definitions.

Chapter 3 - Rings and Modules of Fractions

The notion of $S(\mathfrak{a})$ is particularly important for primary decomposition.

Exercise 3.8 shows that saturated multiplicatively closed subsets of A are in one-to-one correspondence with distinct localizations $S^{-1}A$.

Chapter 4 - Primary Decomposition

?? and ?? are fundamental for understanding primary ideal structure.

Theorem 4.9 is useful for computations.

Chapter 5 - Integral Dependence and Valuations

The most important chapter in the book.

Important results: Lying Over Theorem, Noether Normalization Lemma.

Exercise 5.10 connects the going-up and going-down properties of a ring homomorphism with topological properties of the induced map on spectra, but the ideas are not used further in the book.

Exercise 5.25 uses Jacobson rings to generalize Zariski's Lemma.

Chapter 6 & 7 - Chain Conditions & Noetherian Rings

Most exercises further refine old concepts rather than introducing new ideas.

Chapter 8 - Artinian Rings

An important result is Proposition 8.5.

Exercise 8.3 provides another generalization of Zariski's Lemma.

Chapter 9 - Discrete Valuation Rings and Dedekind Domains

This chapter is closely connected to algebraic number theory. Proposition 9.2 and Theorem 9.3 appear repeatedly throughout the exercises.